

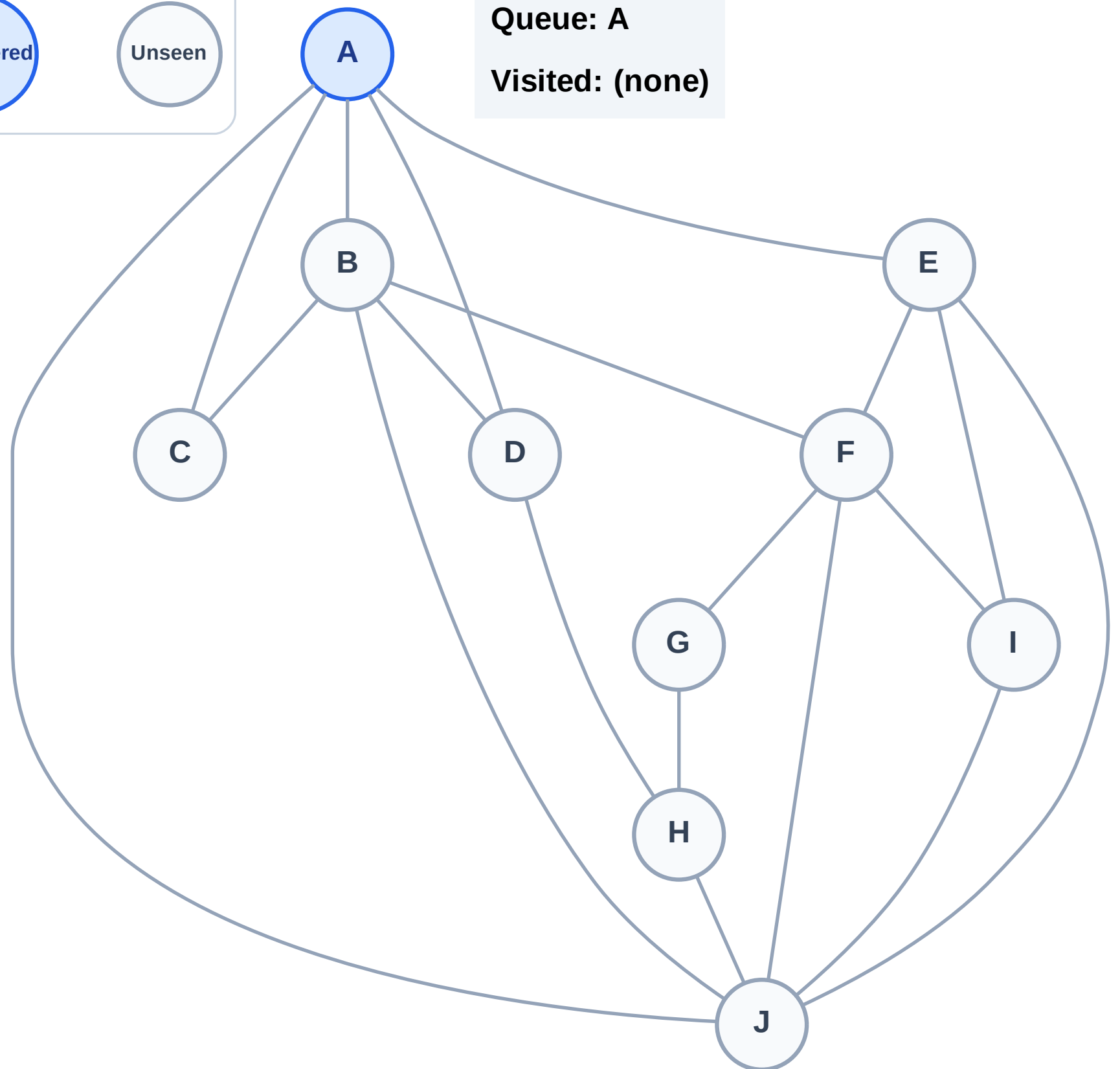
BFS Traversal

Step 1: Start at A

Legend



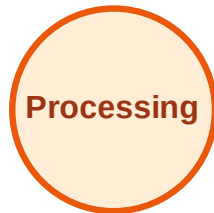
Queue: A
Visited: (none)



BFS Traversal

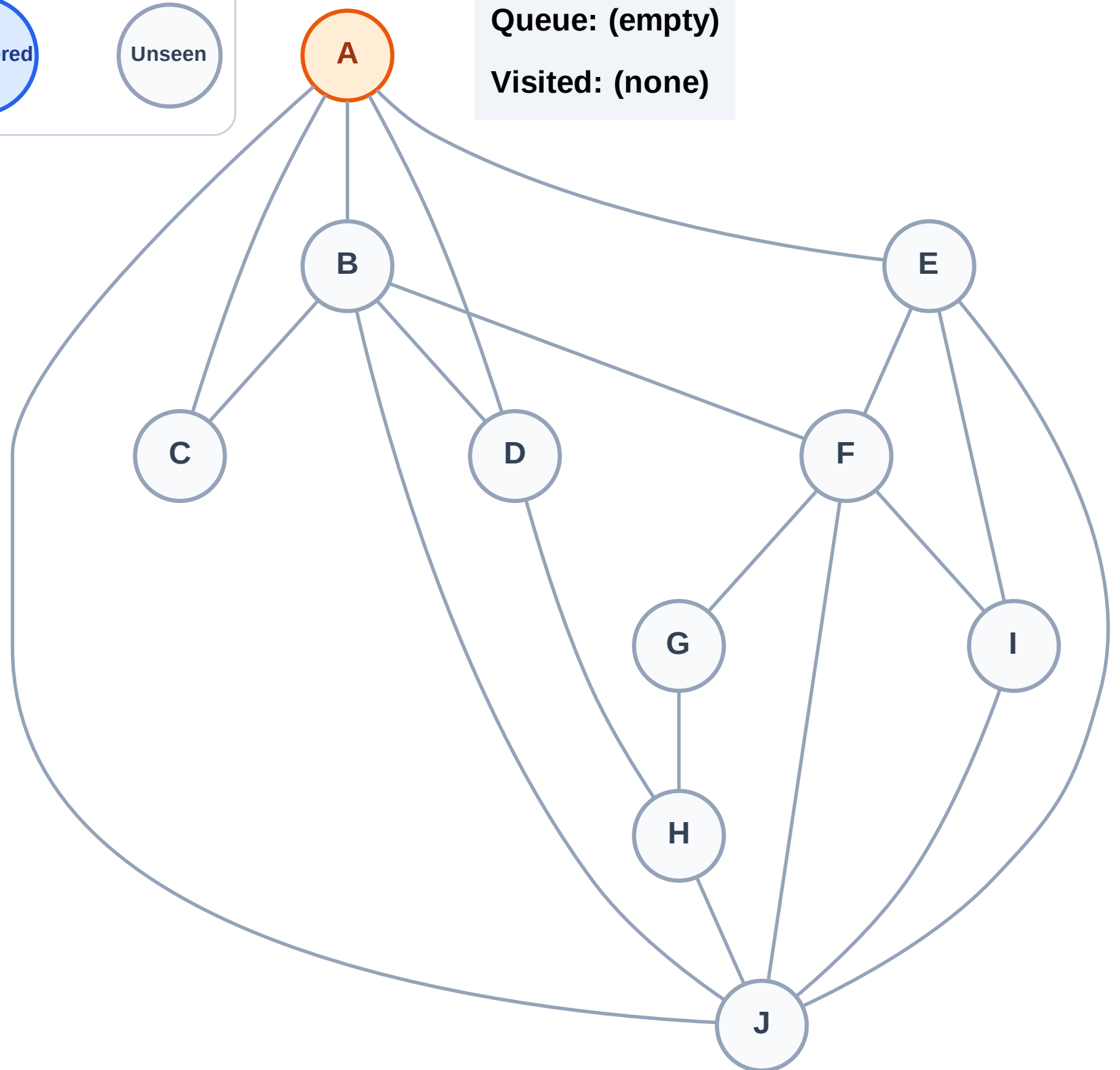
Step 2: Process node A

Legend



Queue: (empty)

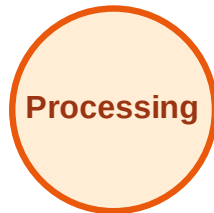
Visited: (none)



BFS Traversal

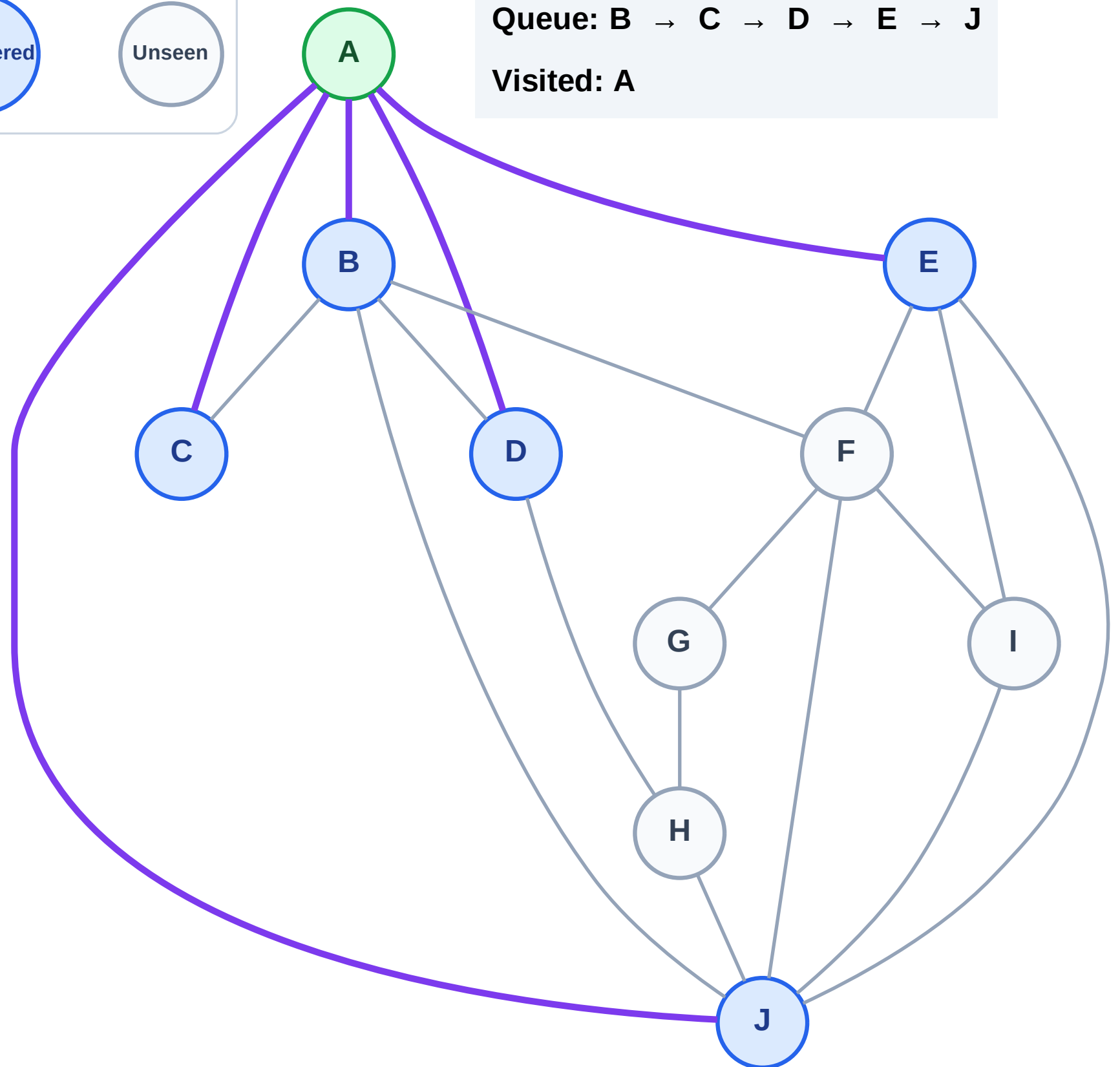
Step 3: Discover B, C, D, E, J from A

Legend



Queue: B → C → D → E → J

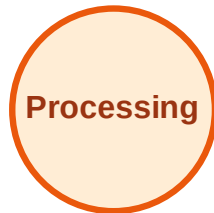
Visited: A



BFS Traversal

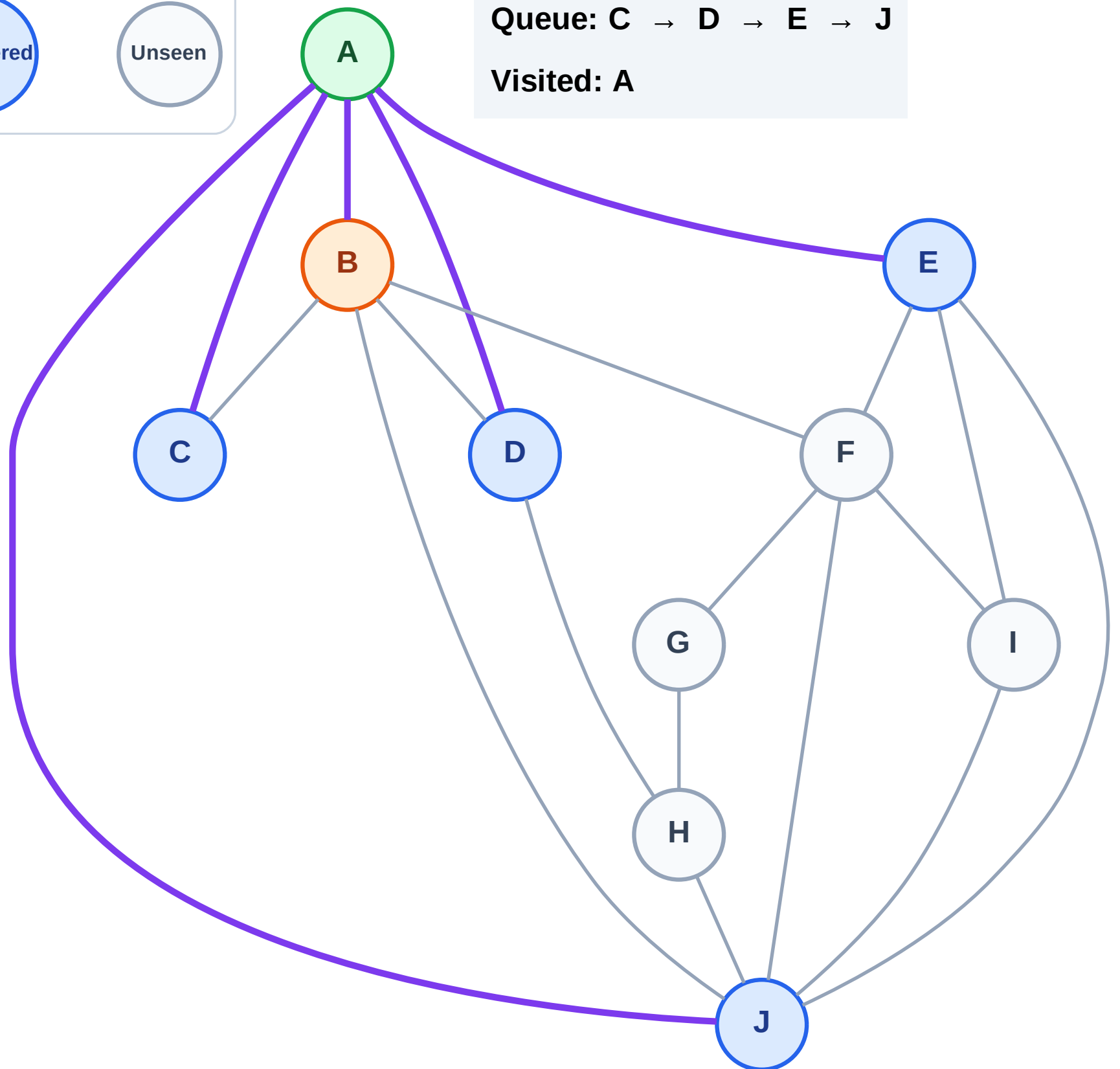
Step 4: Process node B

Legend



Queue: C → D → E → J

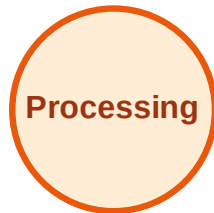
Visited: A



BFS Traversal

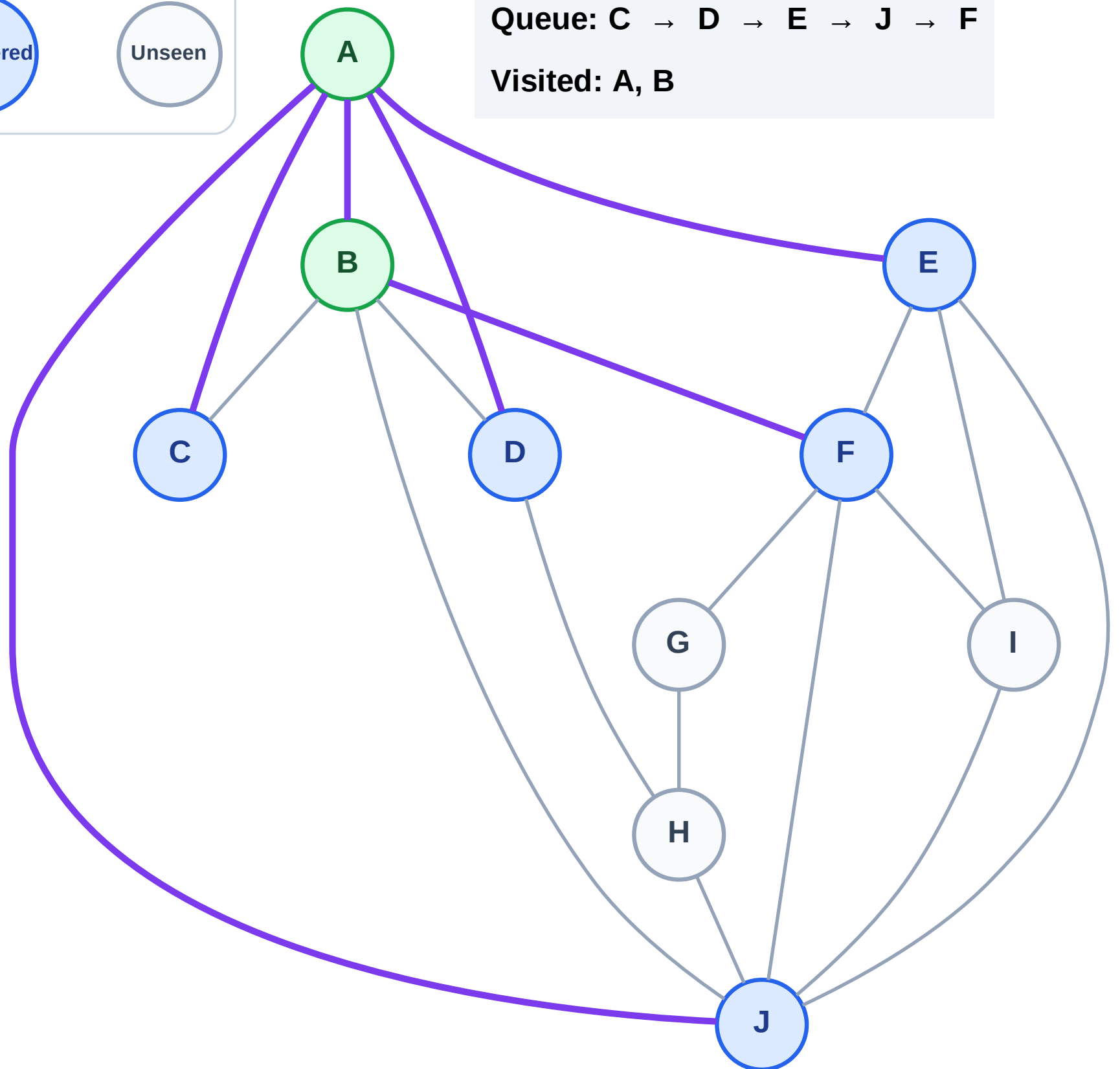
Step 5: Discover F from B

Legend



Queue: C → D → E → J → F

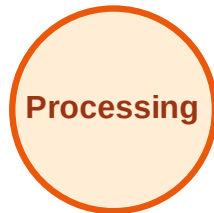
Visited: A, B



BFS Traversal

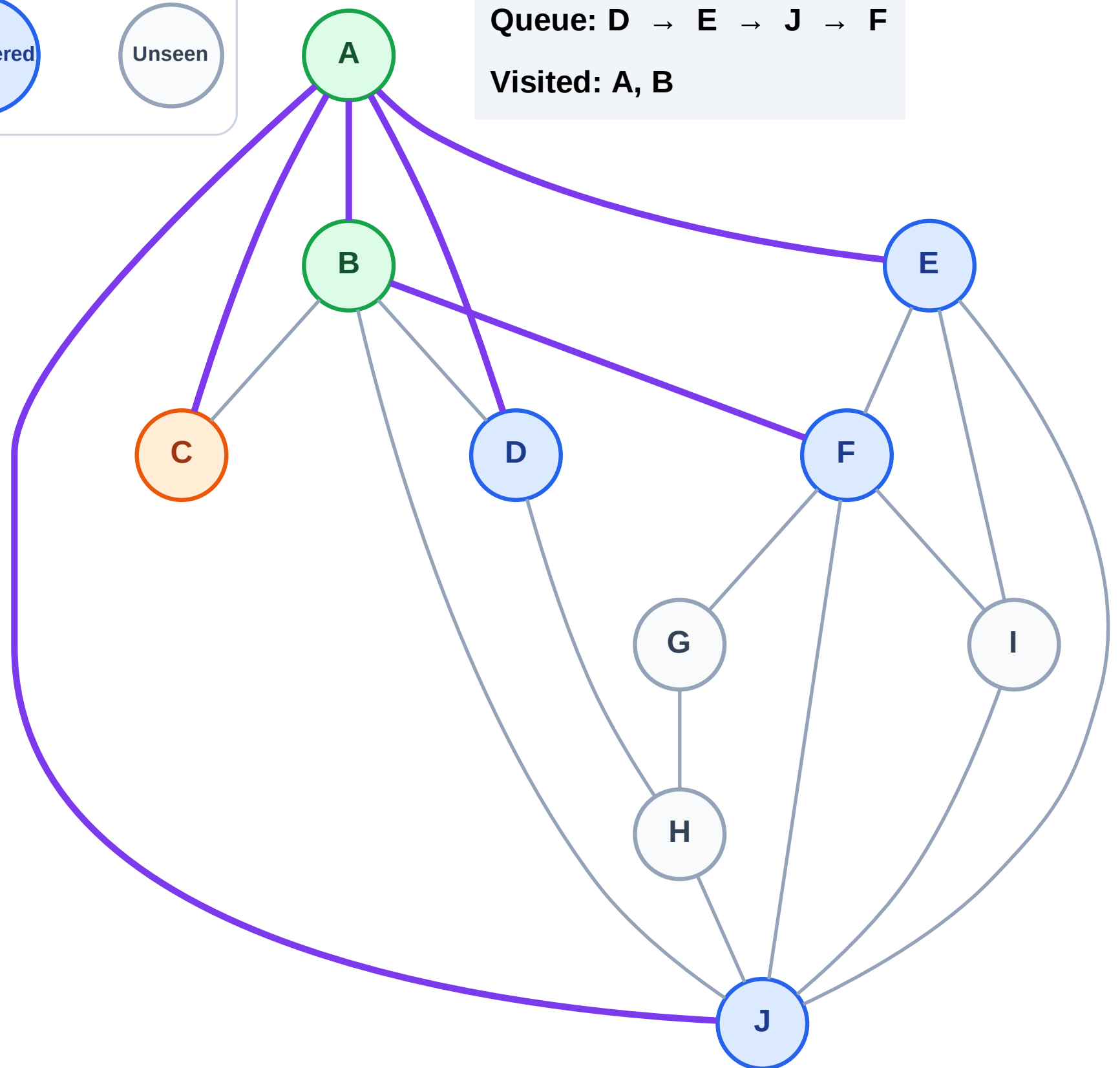
Step 6: Process node C

Legend



Queue: D → E → J → F

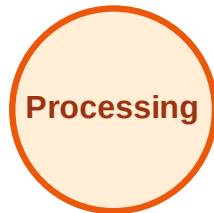
Visited: A, B



BFS Traversal

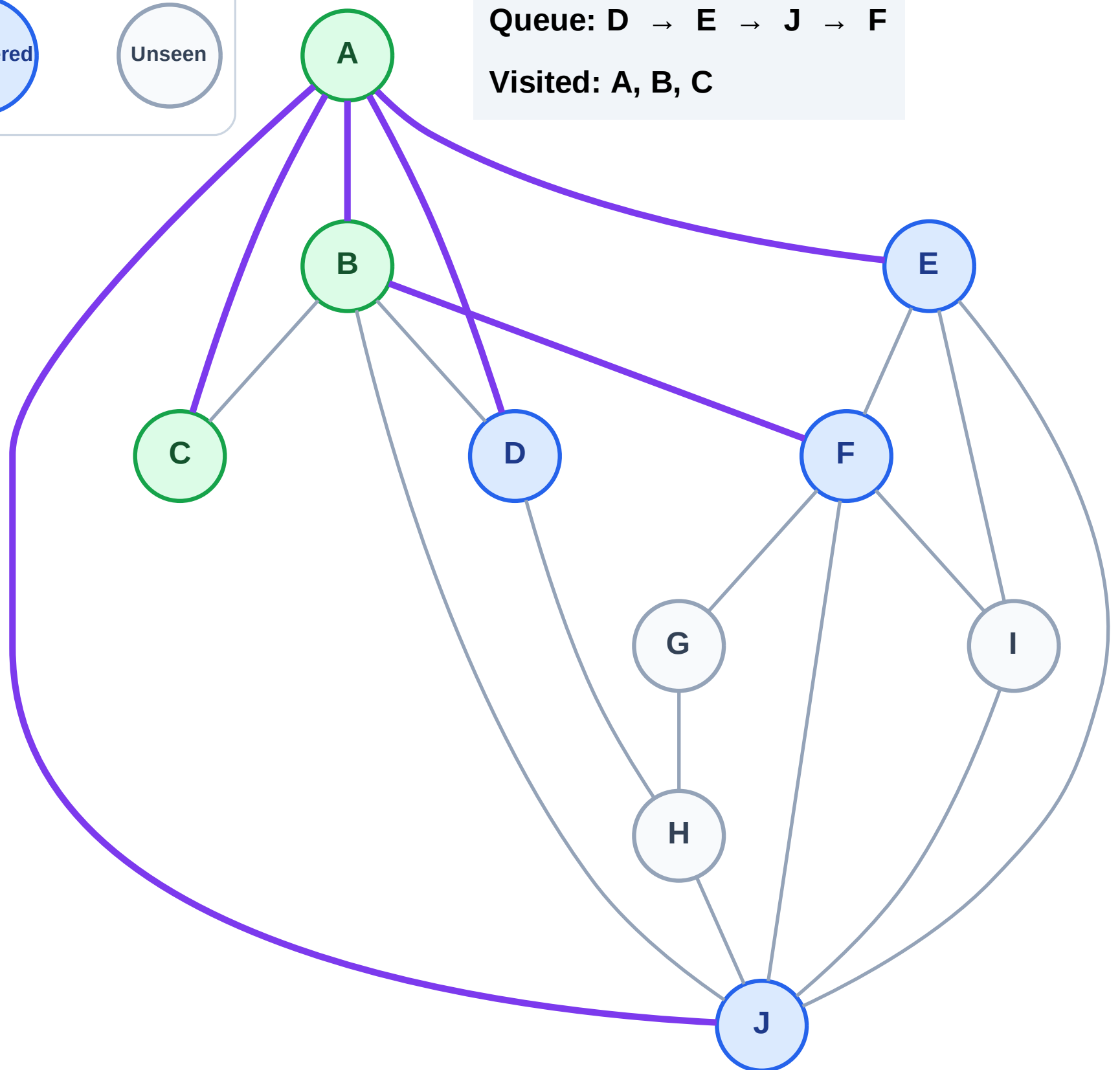
Step 7: No new neighbors from C

Legend



Queue: D → E → J → F

Visited: A, B, C



BFS Traversal

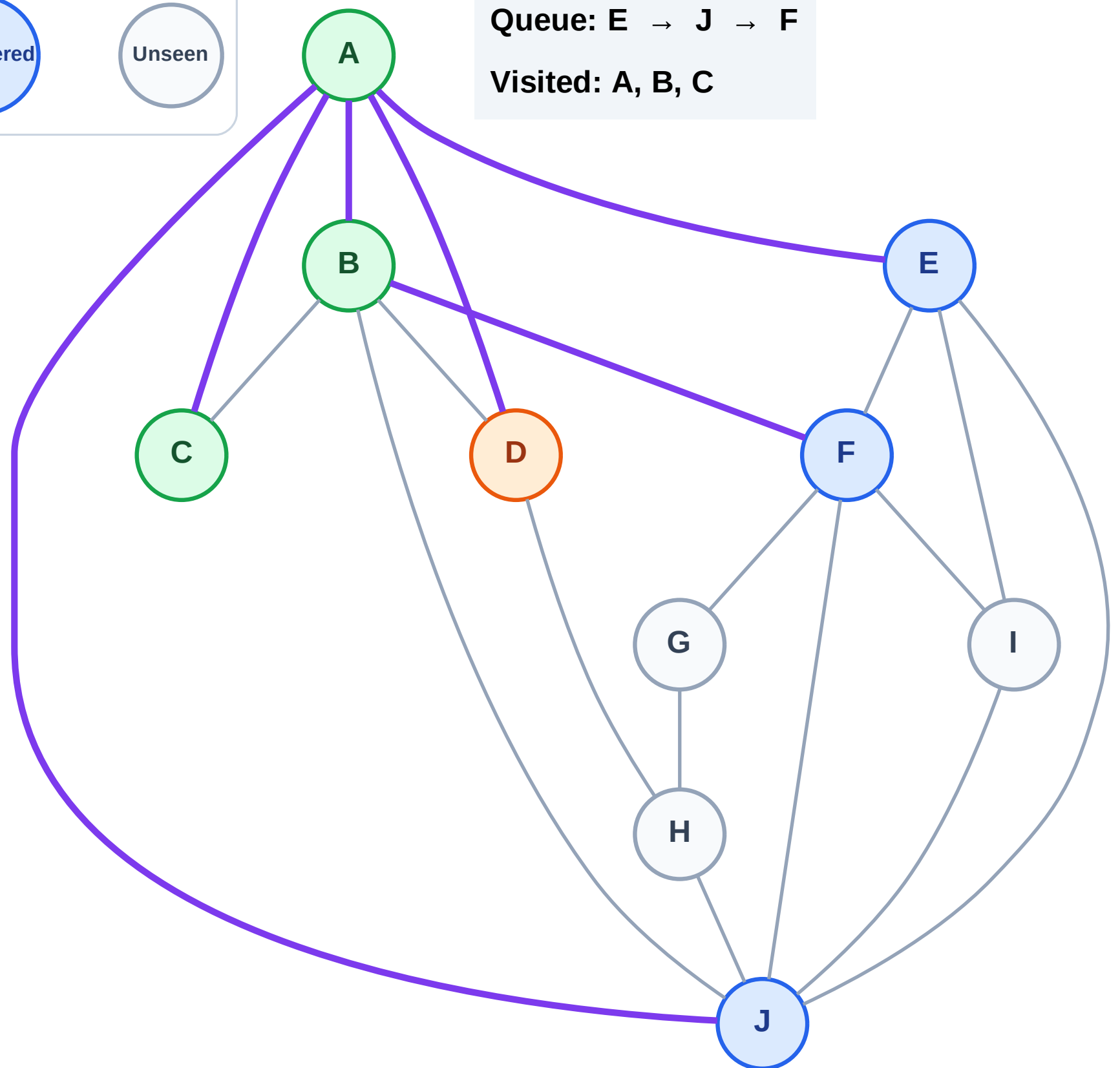
Step 8: Process node D

Legend



Queue: E → J → F

Visited: A, B, C



BFS Traversal

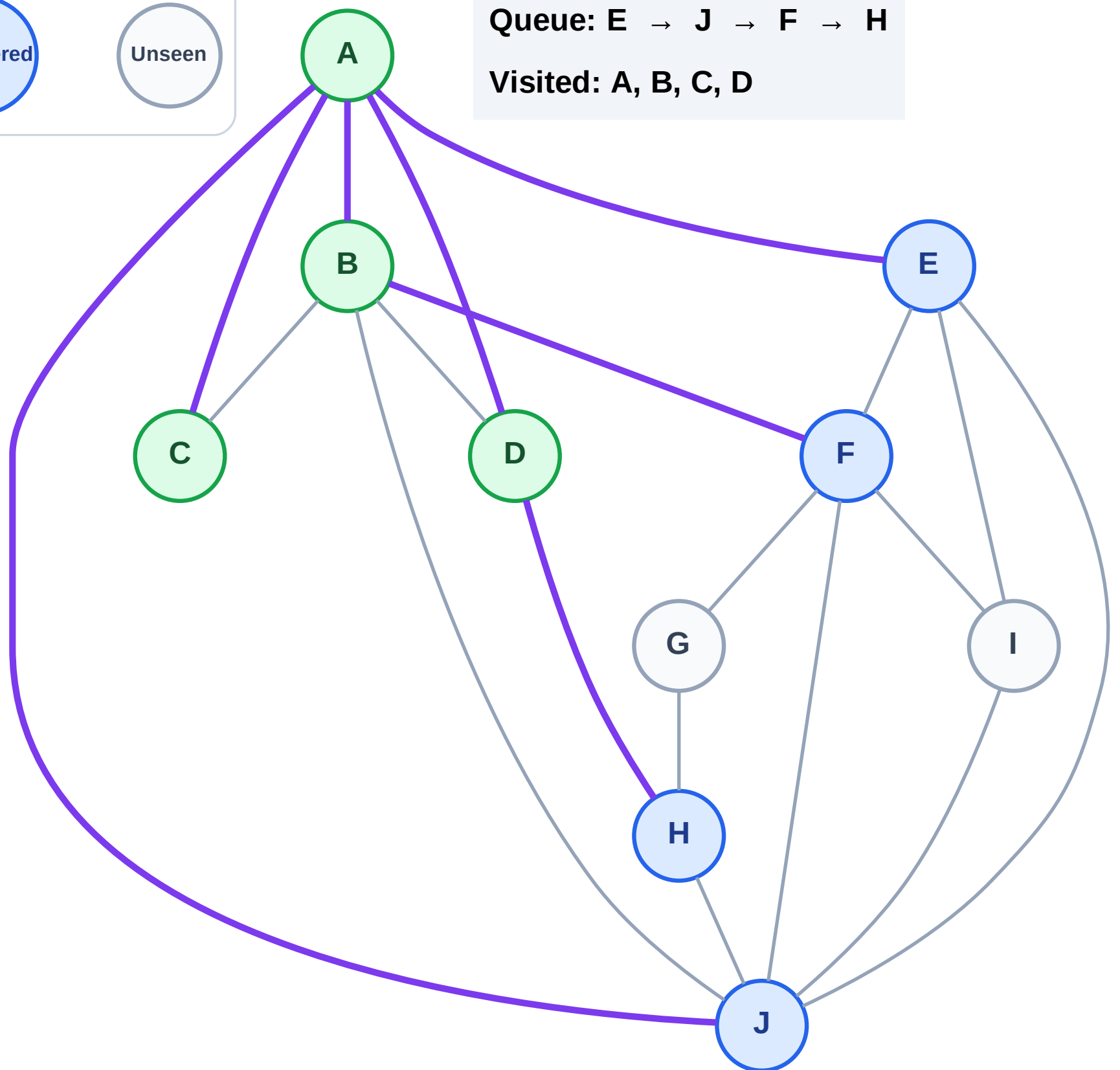
Step 9: Discover H from D

Legend



Queue: E → J → F → H

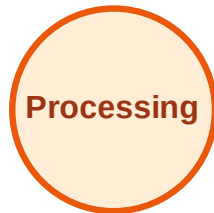
Visited: A, B, C, D



BFS Traversal

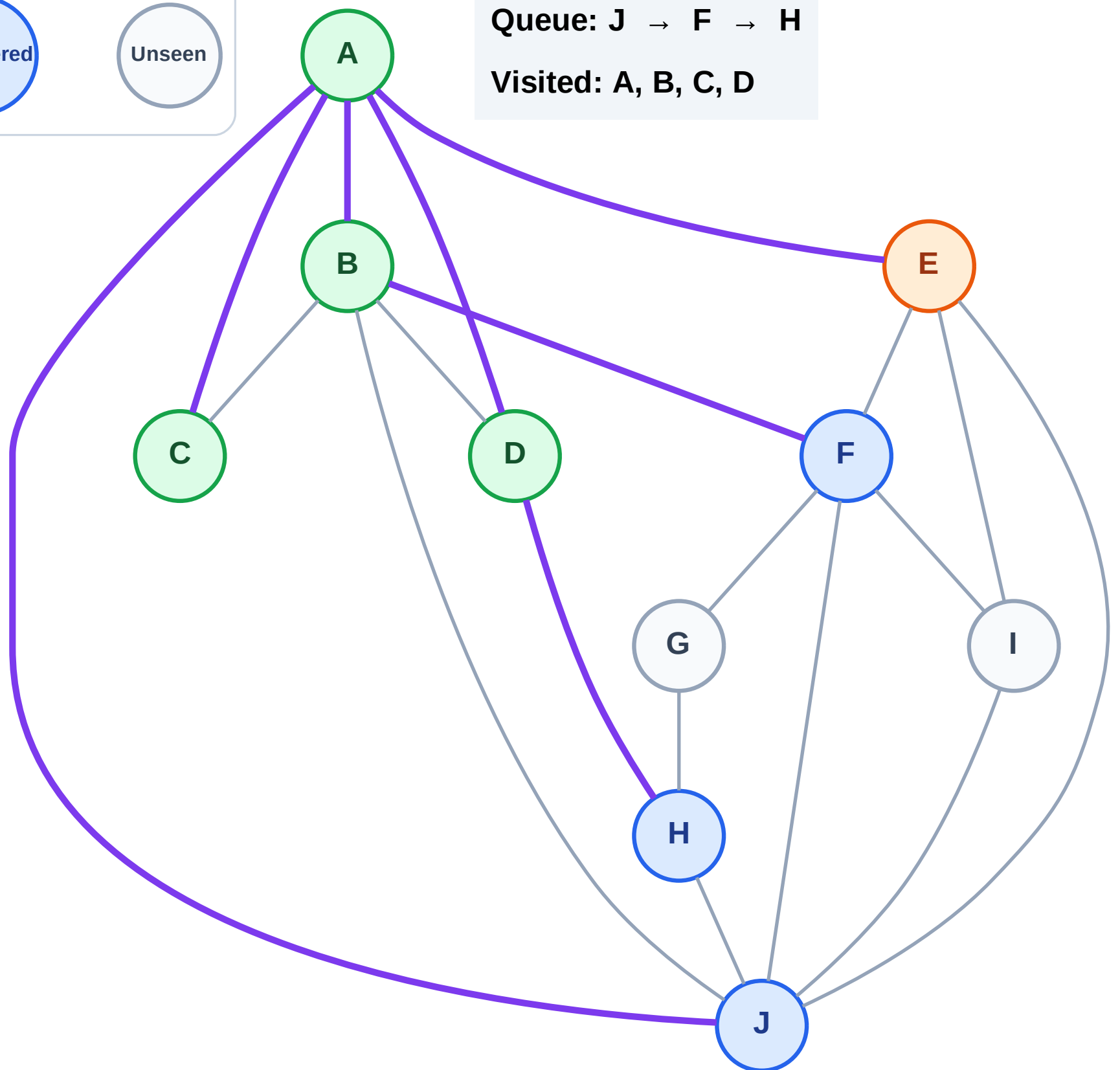
Step 10: Process node E

Legend



Queue: J → F → H

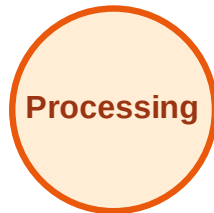
Visited: A, B, C, D



BFS Traversal

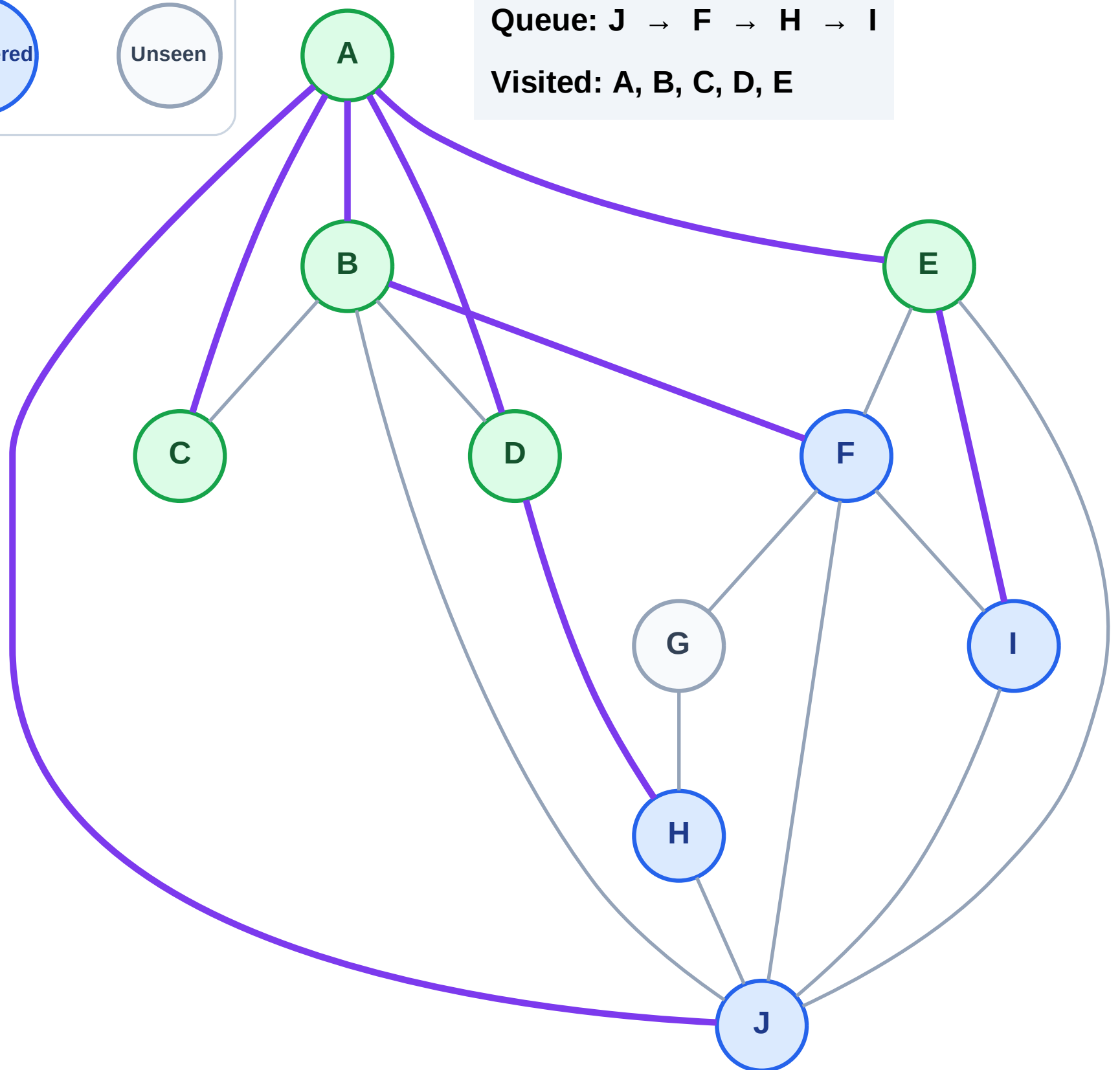
Step 11: Discover I from E

Legend



Queue: J → F → H → I

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node J

Legend

Complete

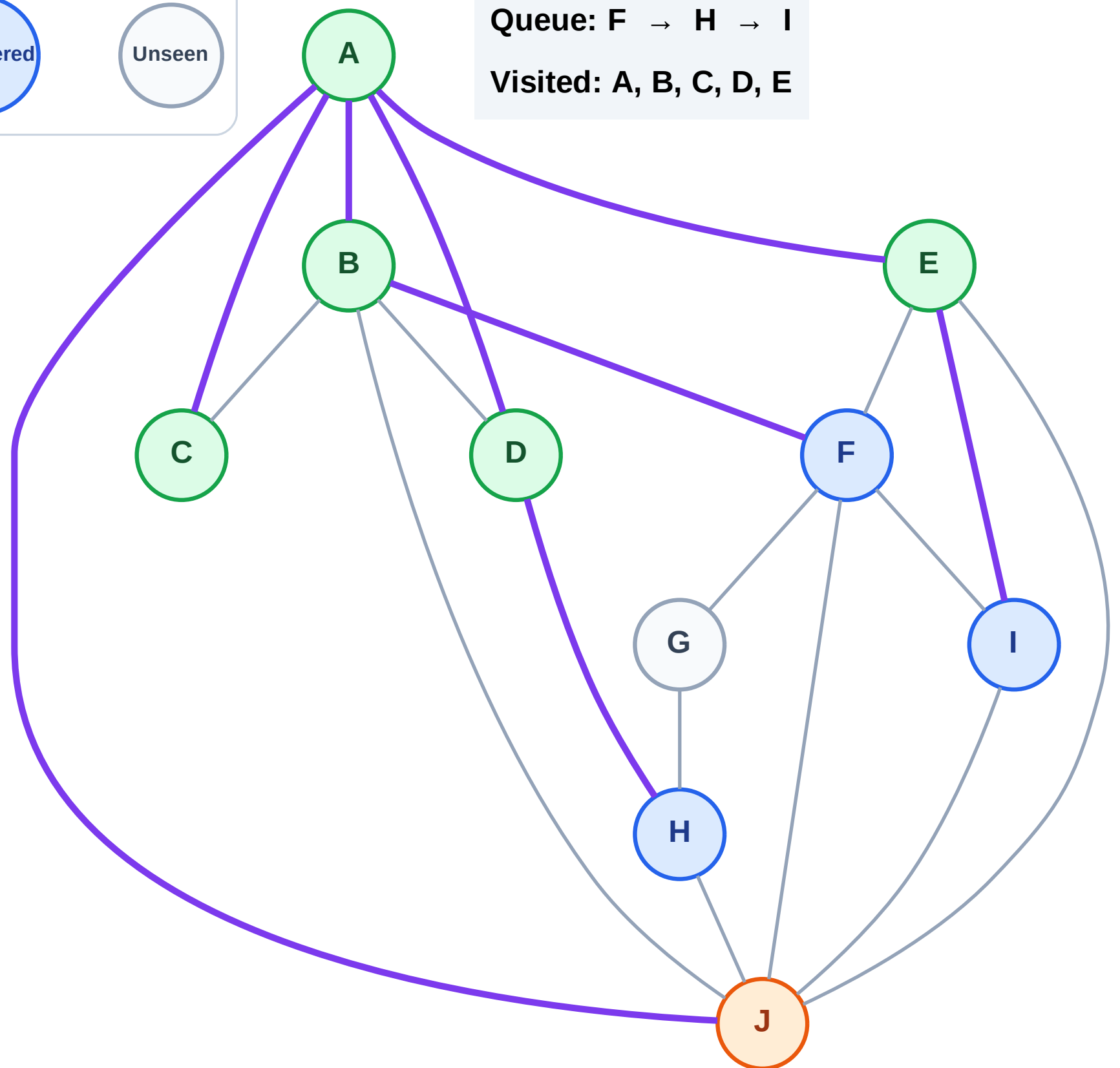
Processing

Discovered

Unseen

Queue: F → H → I

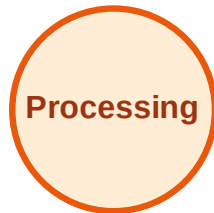
Visited: A, B, C, D, E



BFS Traversal

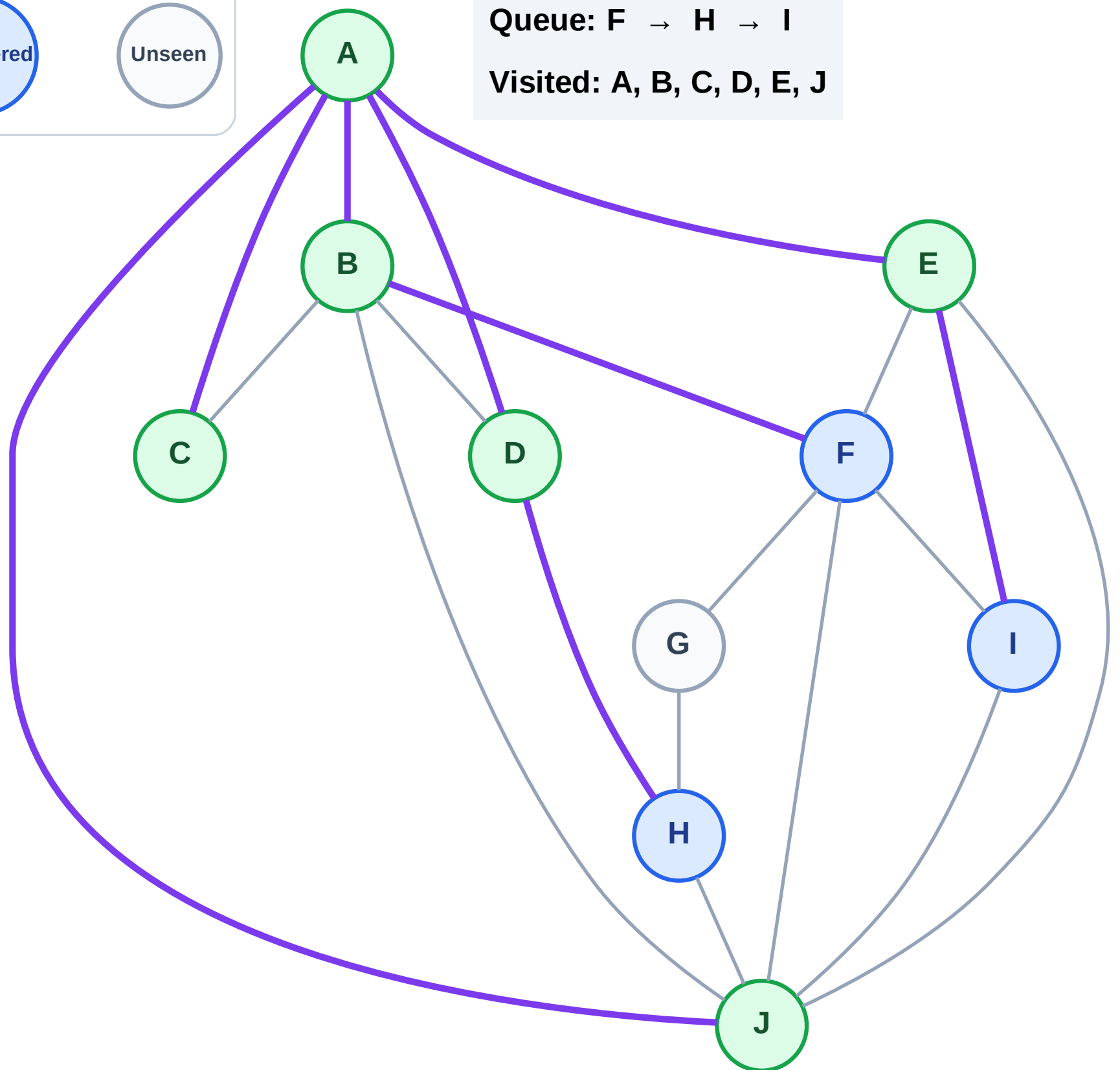
Step 13: No new neighbors from J

Legend



Queue: F → H → I

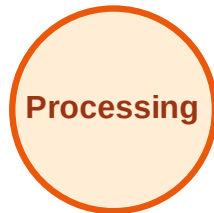
Visited: A, B, C, D, E, J



BFS Traversal

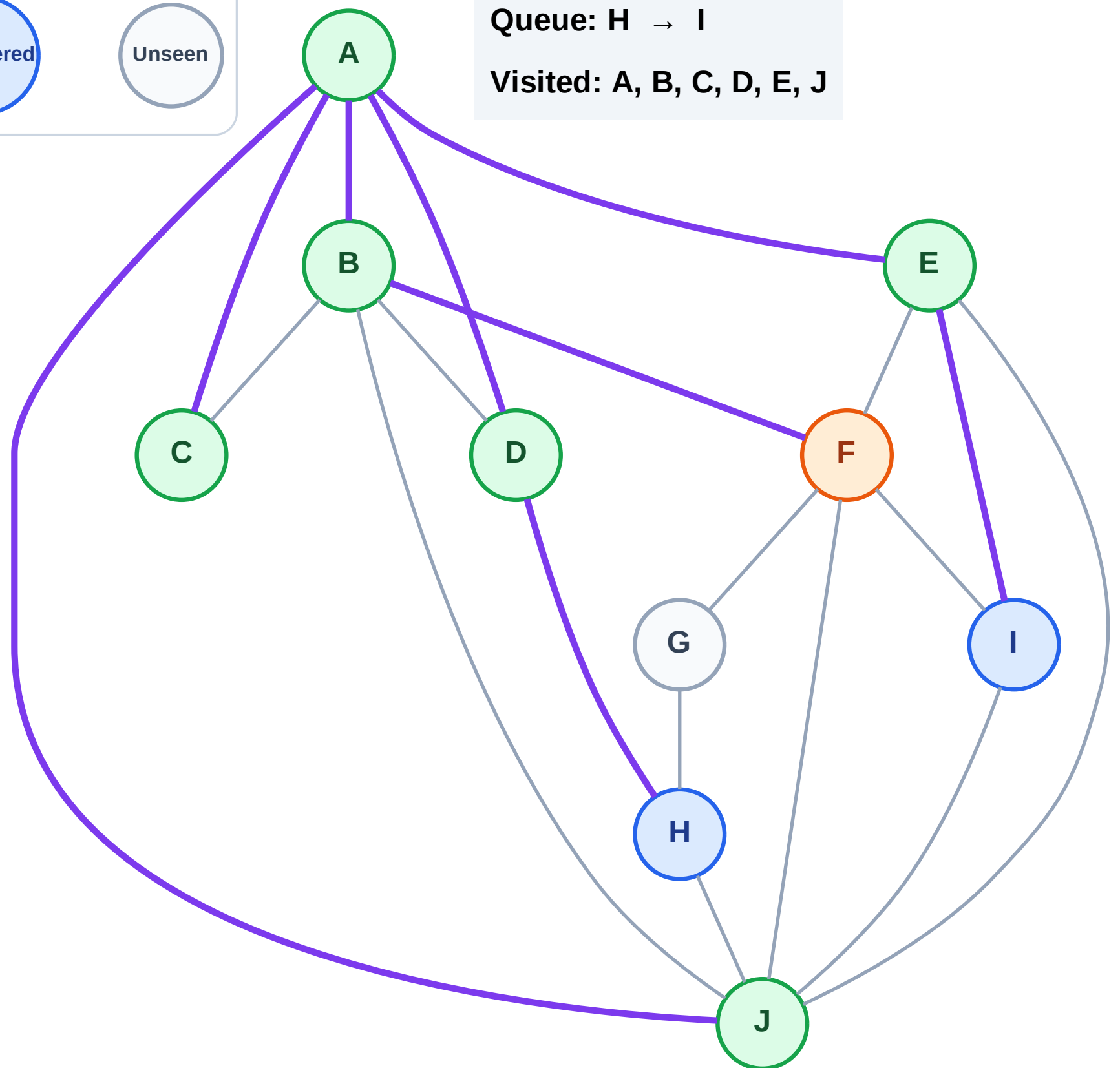
Step 14: Process node F

Legend



Queue: H → I

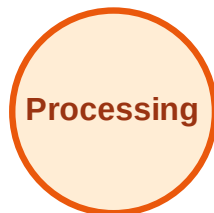
Visited: A, B, C, D, E, J



BFS Traversal

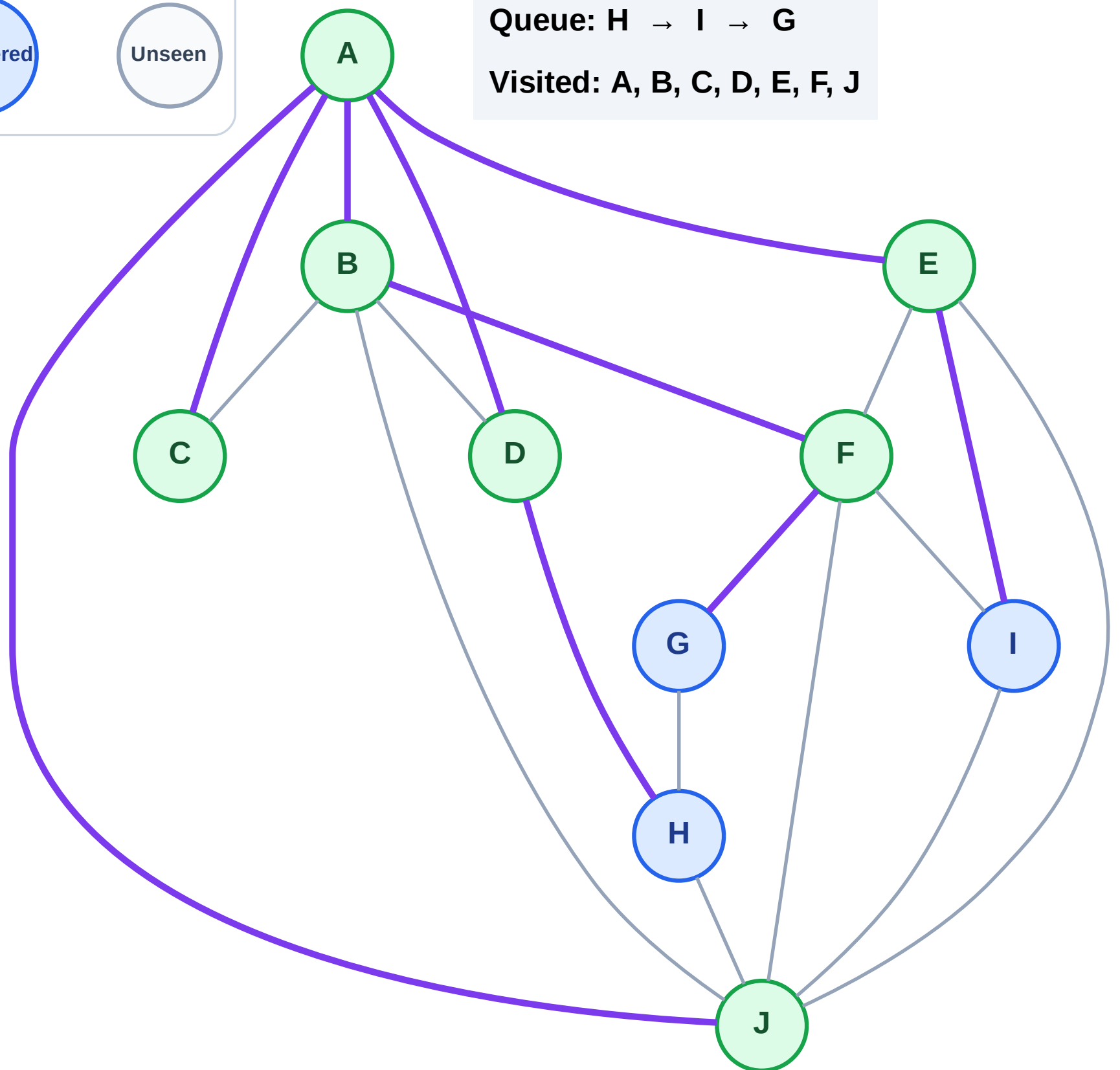
Step 15: Discover G from F

Legend



Queue: H → I → G

Visited: A, B, C, D, E, F, J



BFS Traversal

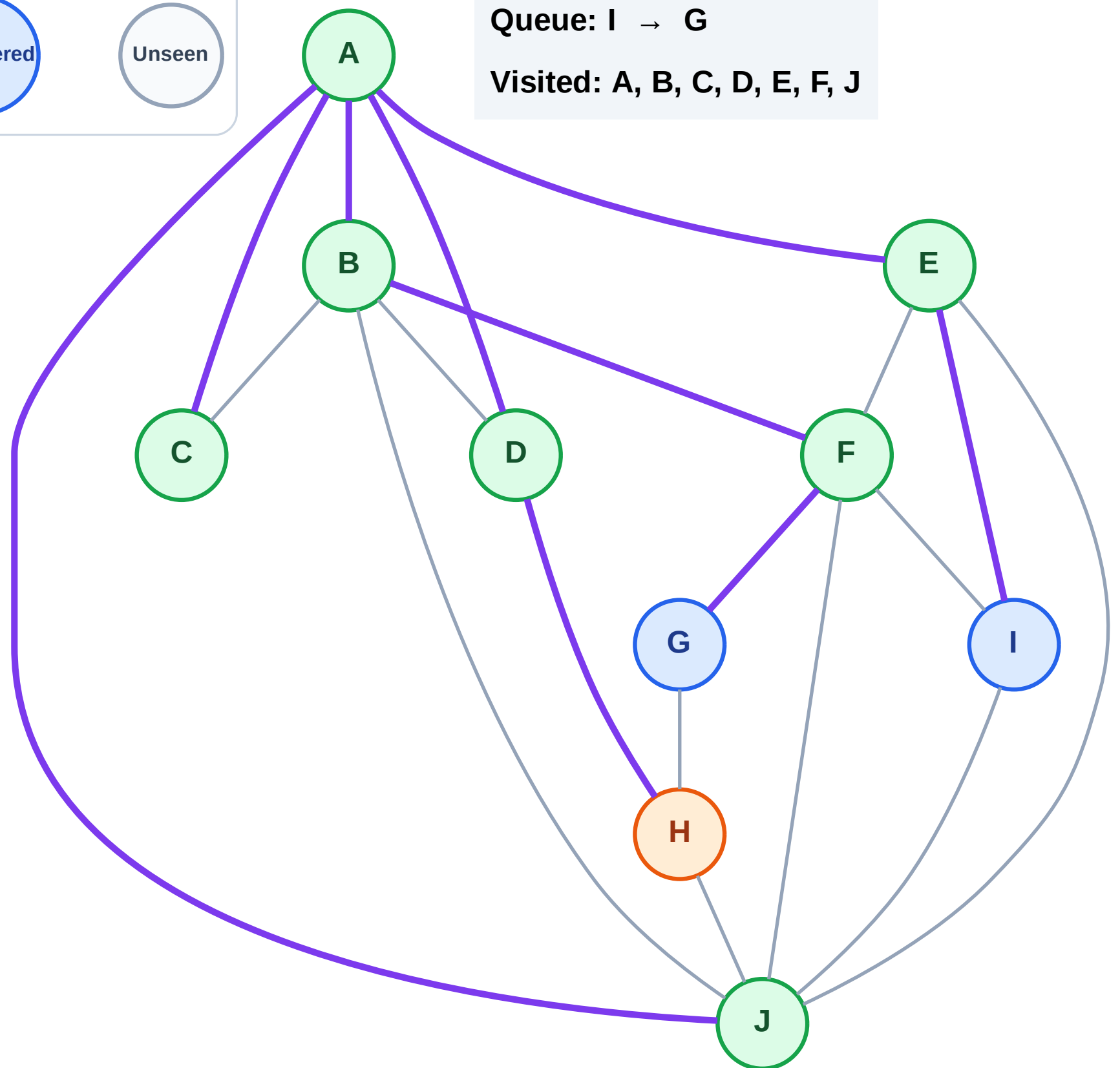
Step 16: Process node H

Legend



Queue: I → G

Visited: A, B, C, D, E, F, J



BFS Traversal

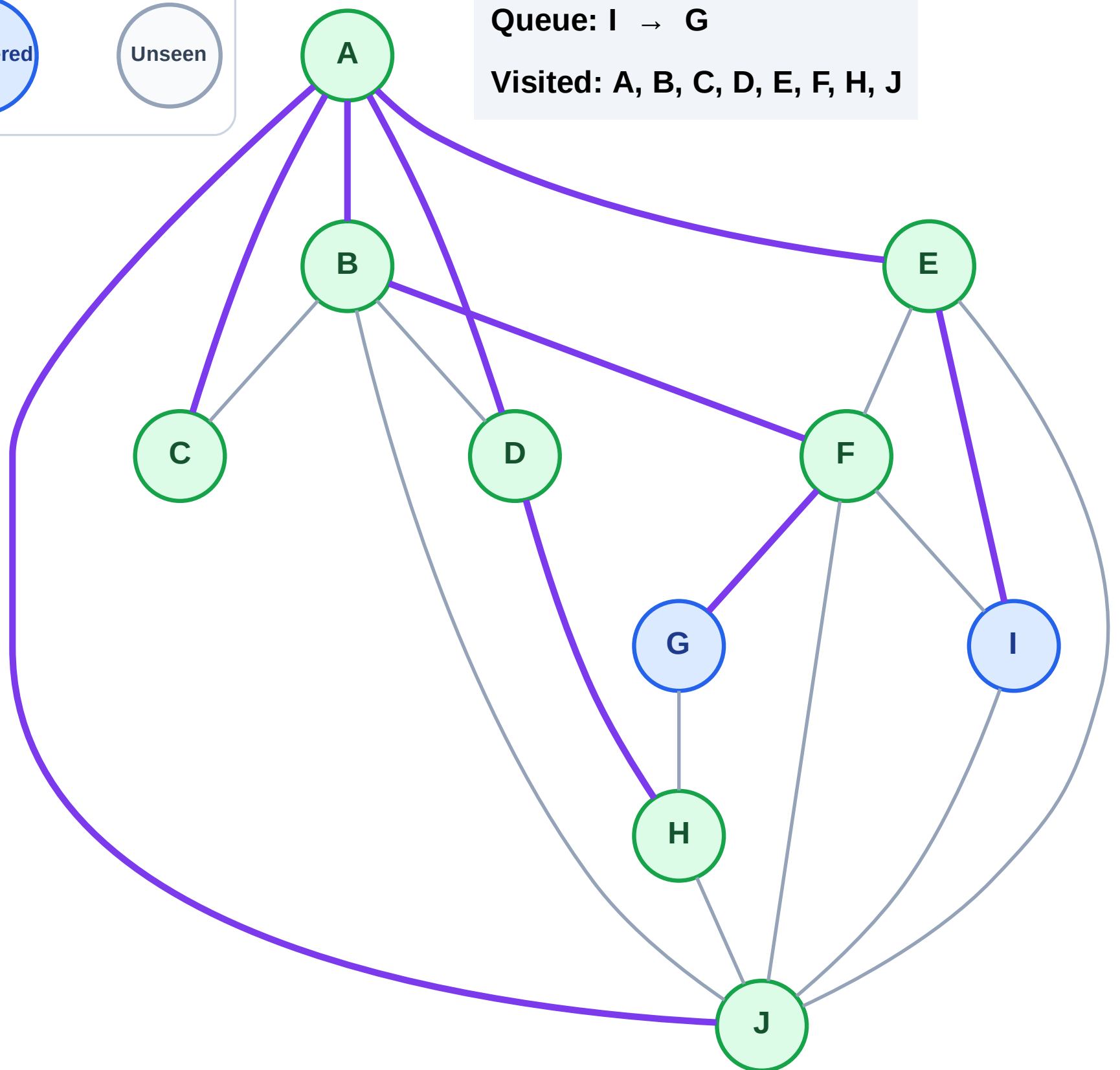
Step 17: No new neighbors from H

Legend



Queue: I → G

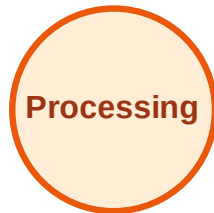
Visited: A, B, C, D, E, F, H, J



BFS Traversal

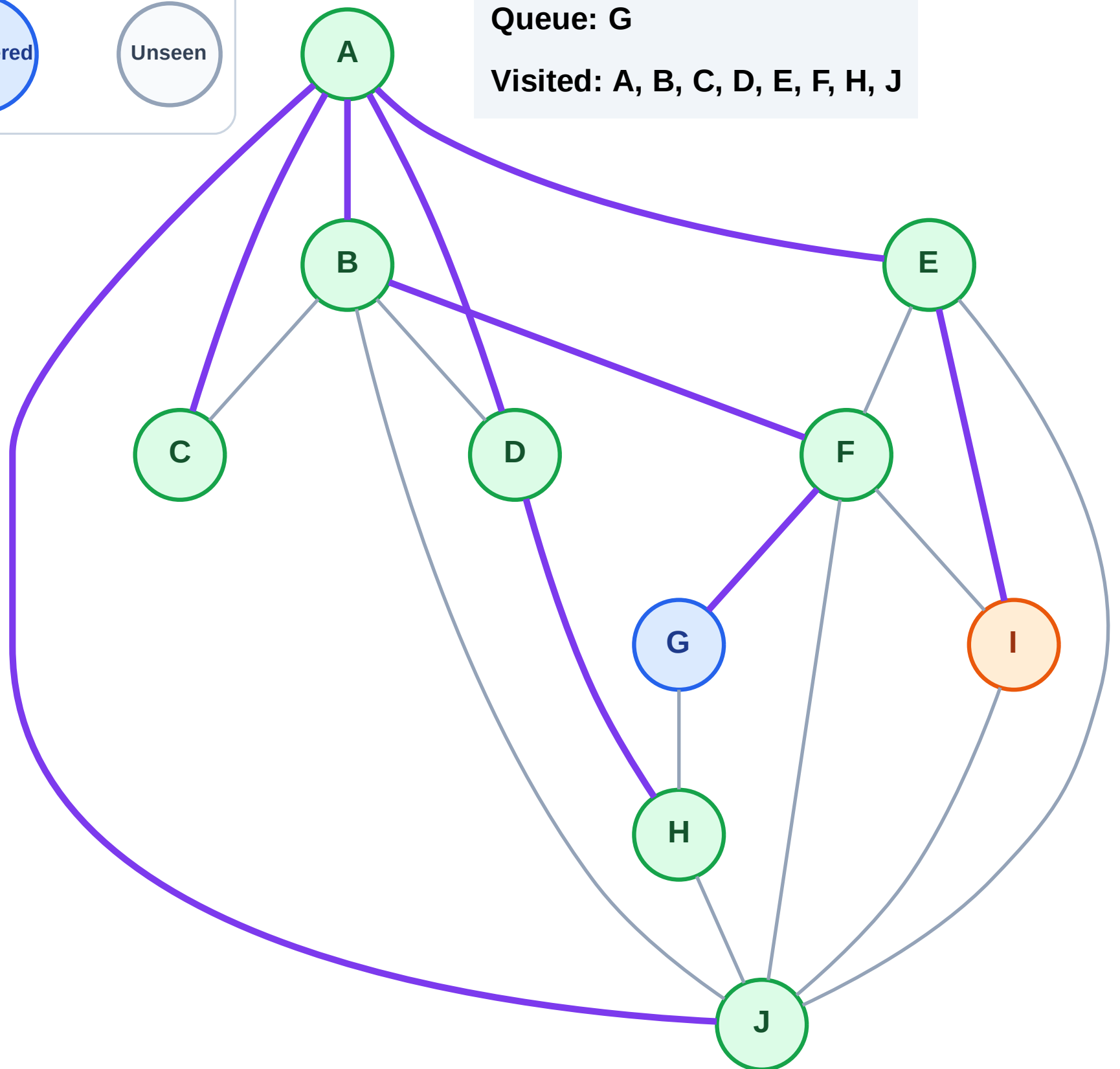
Step 18: Process node I

Legend



Queue: G

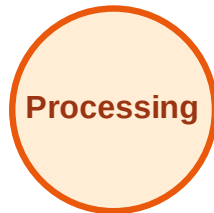
Visited: A, B, C, D, E, F, H, J



BFS Traversal

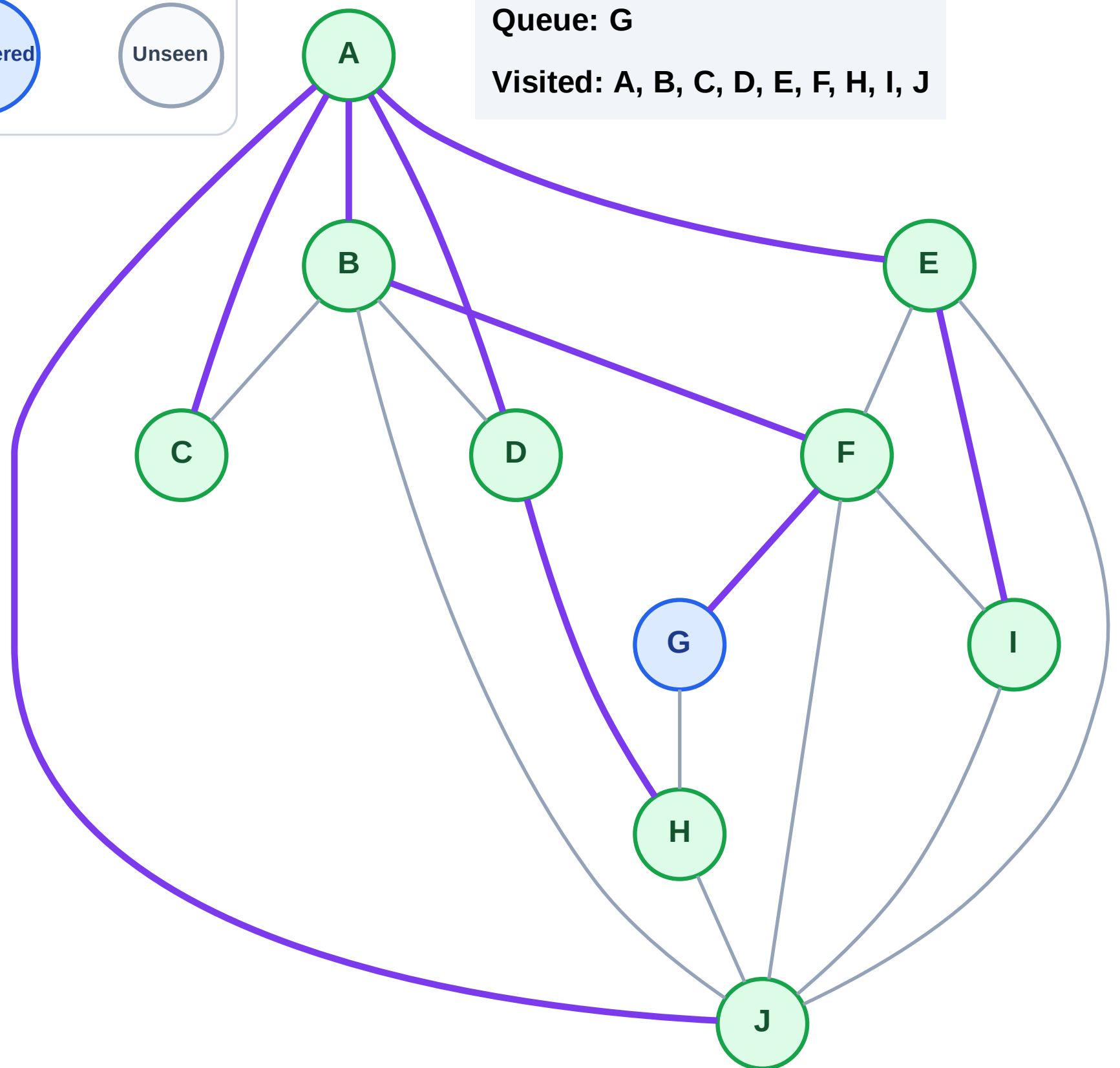
Step 19: No new neighbors from I

Legend



Queue: G

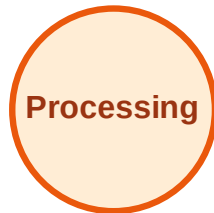
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

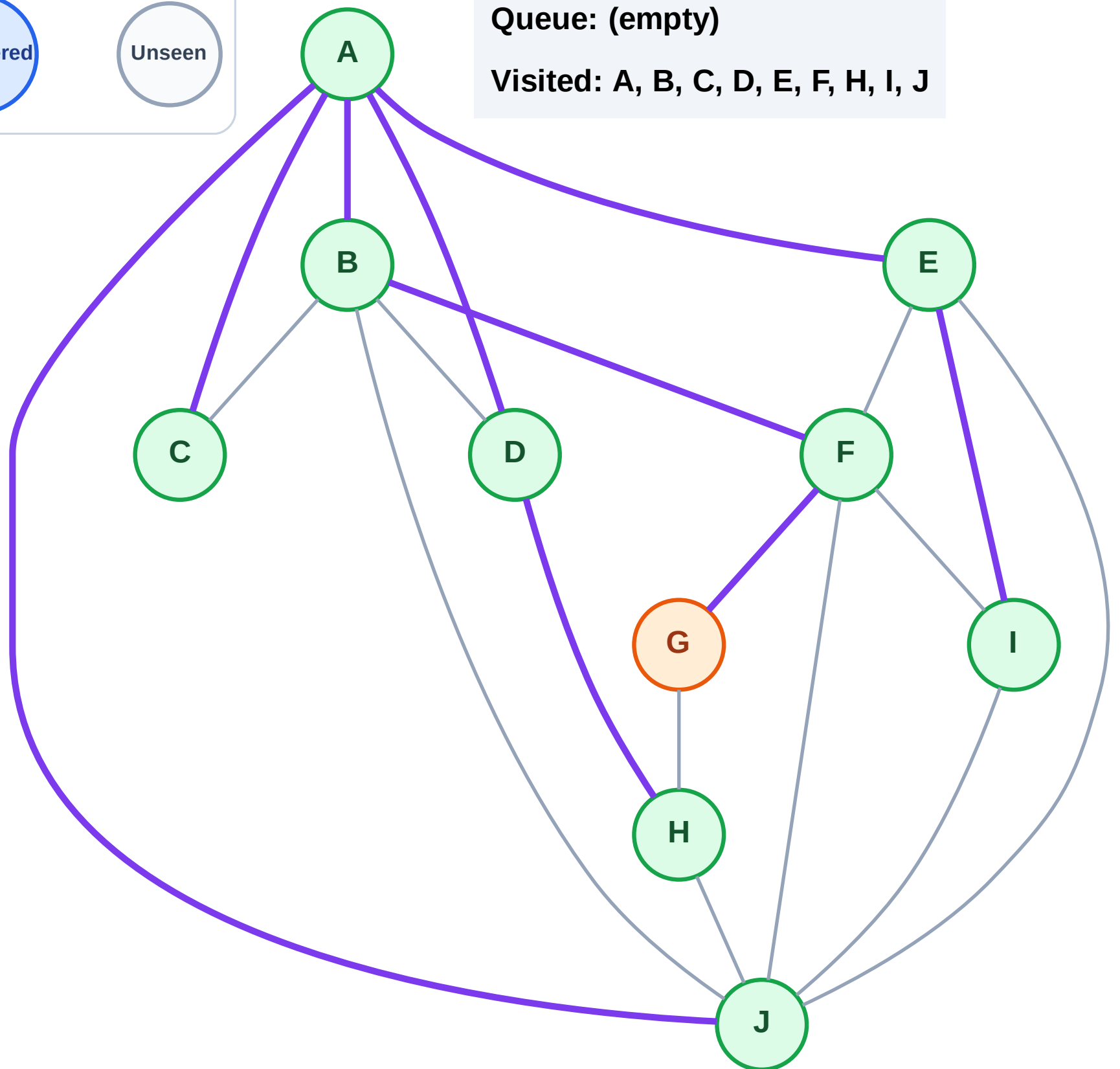
Step 20: Process node G

Legend



Queue: (empty)

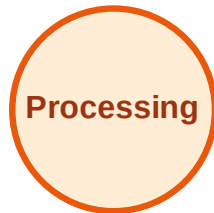
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

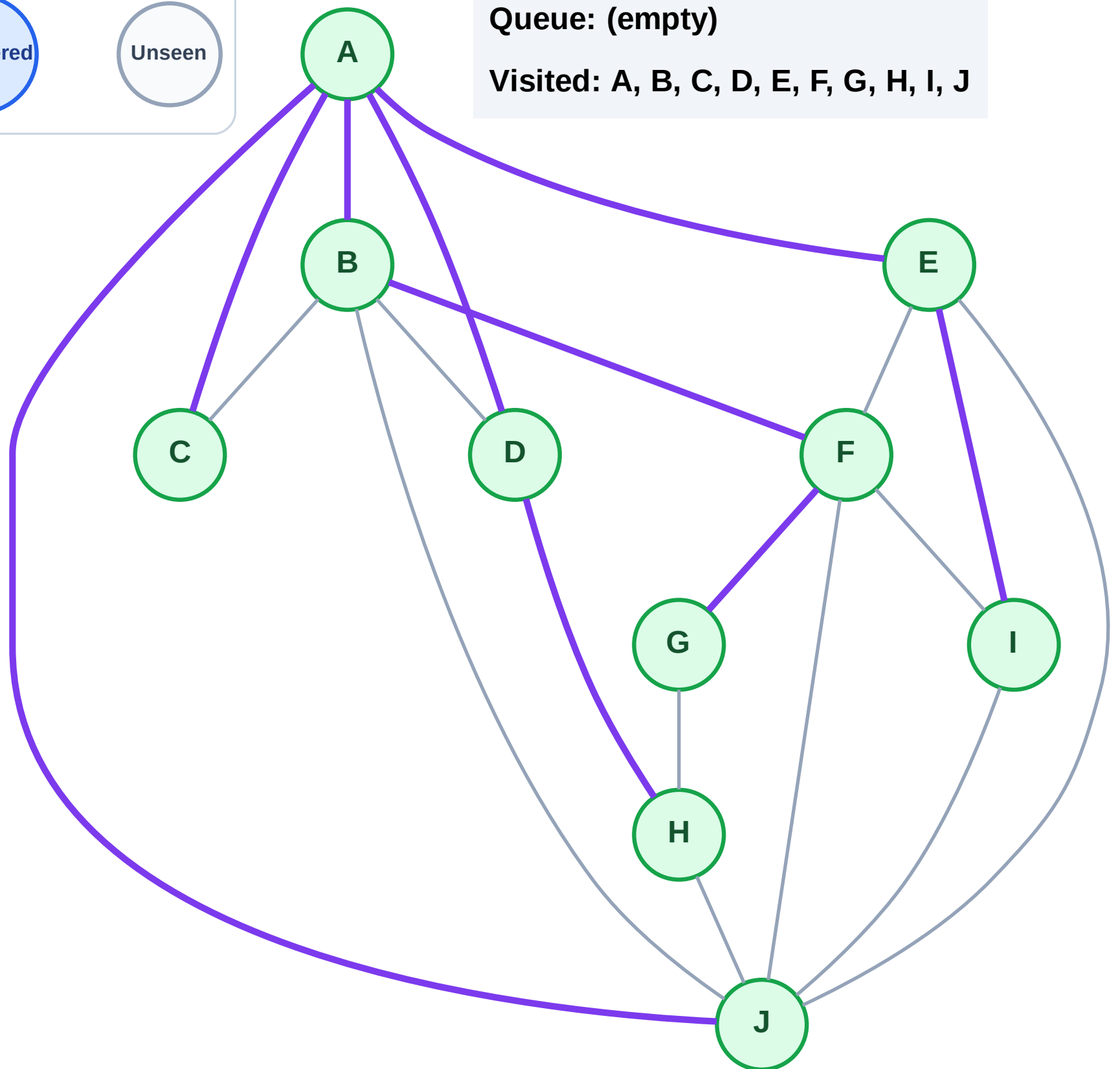
Step 21: No new neighbors from G

Legend



Queue: (empty)

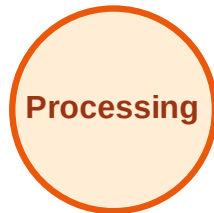
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

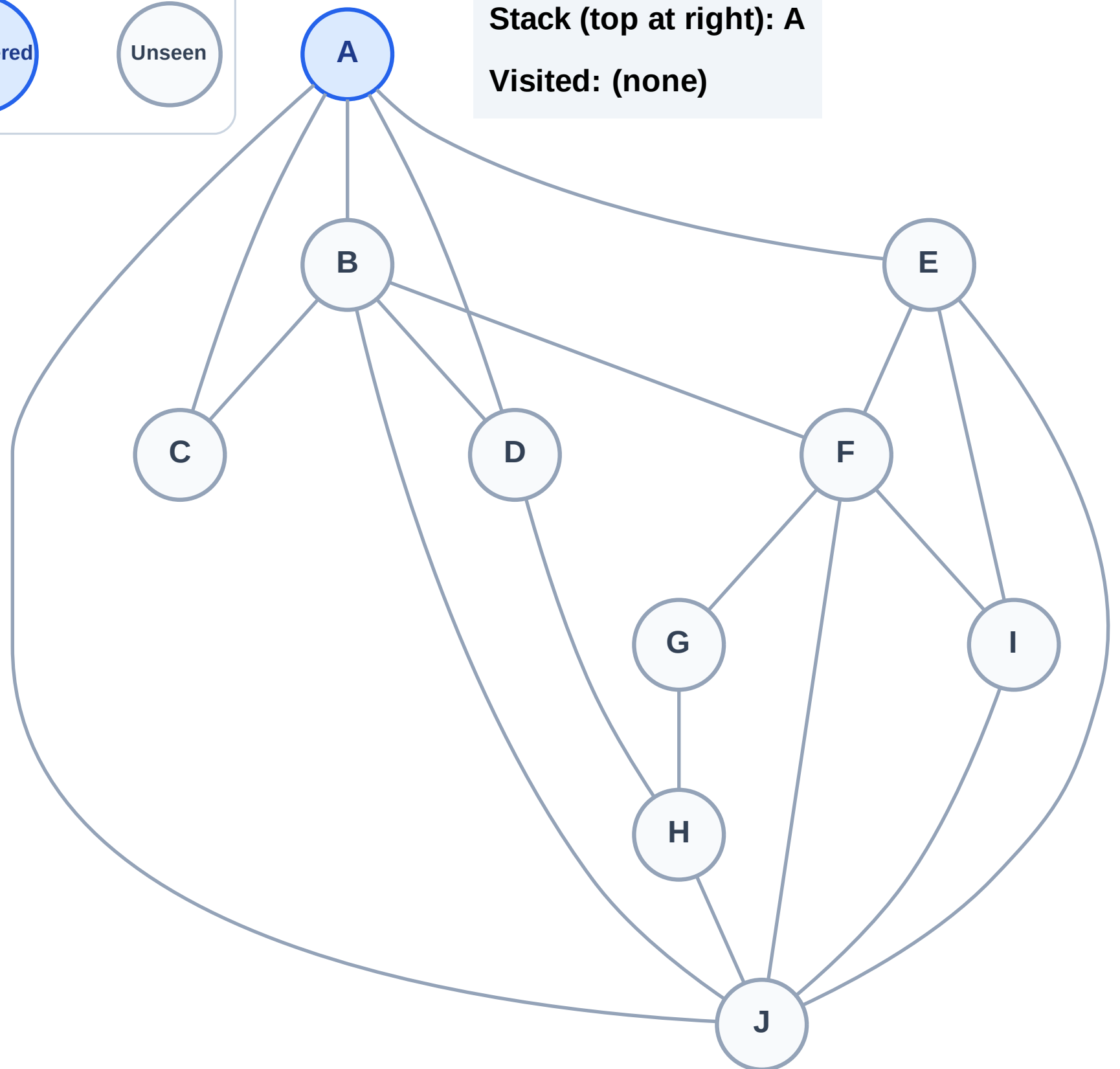
Step 1: Start at A

Legend



Stack (top at right): A

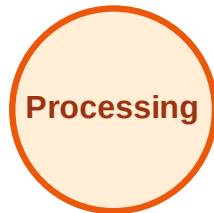
Visited: (none)



DFS Traversal

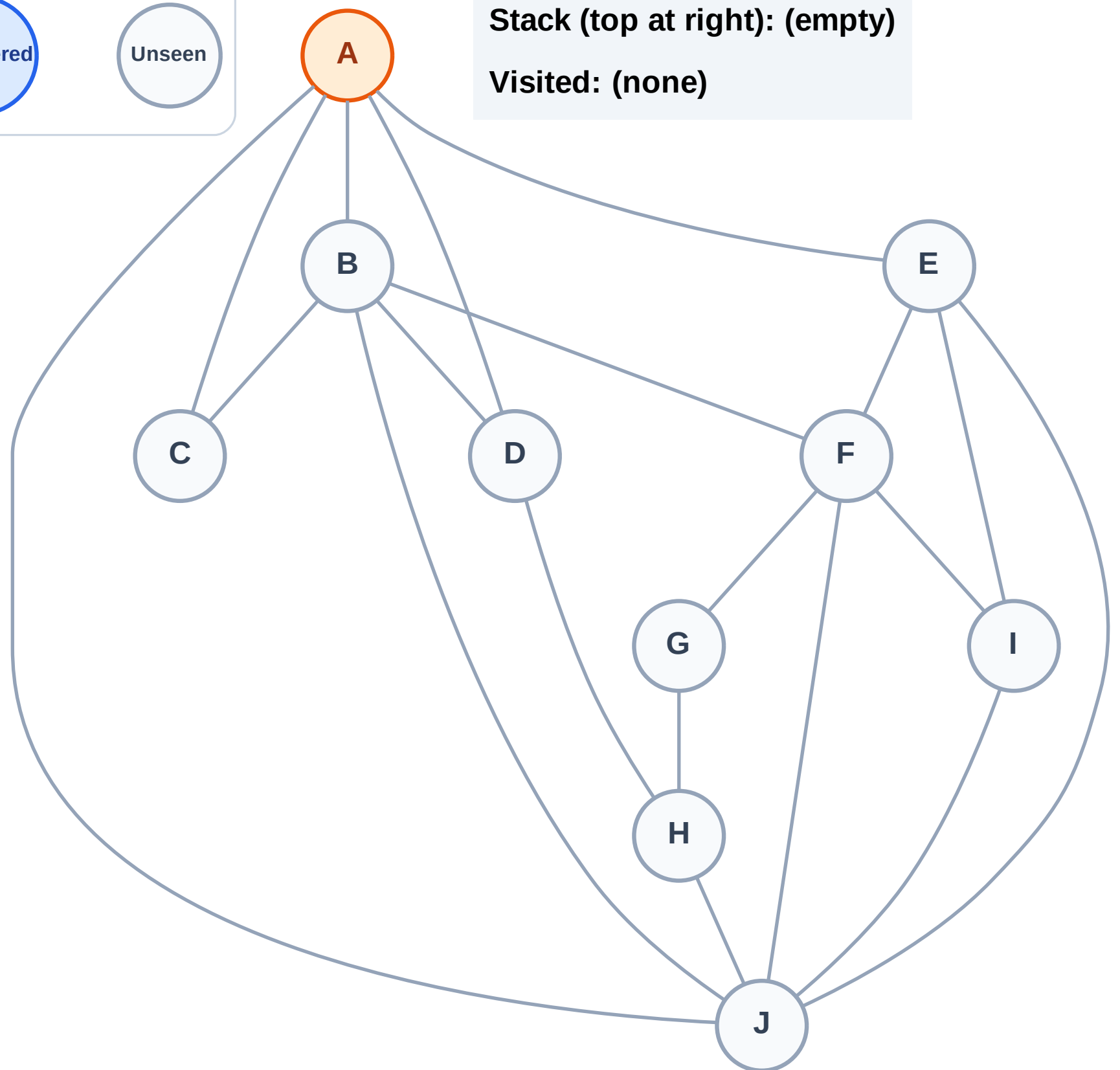
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

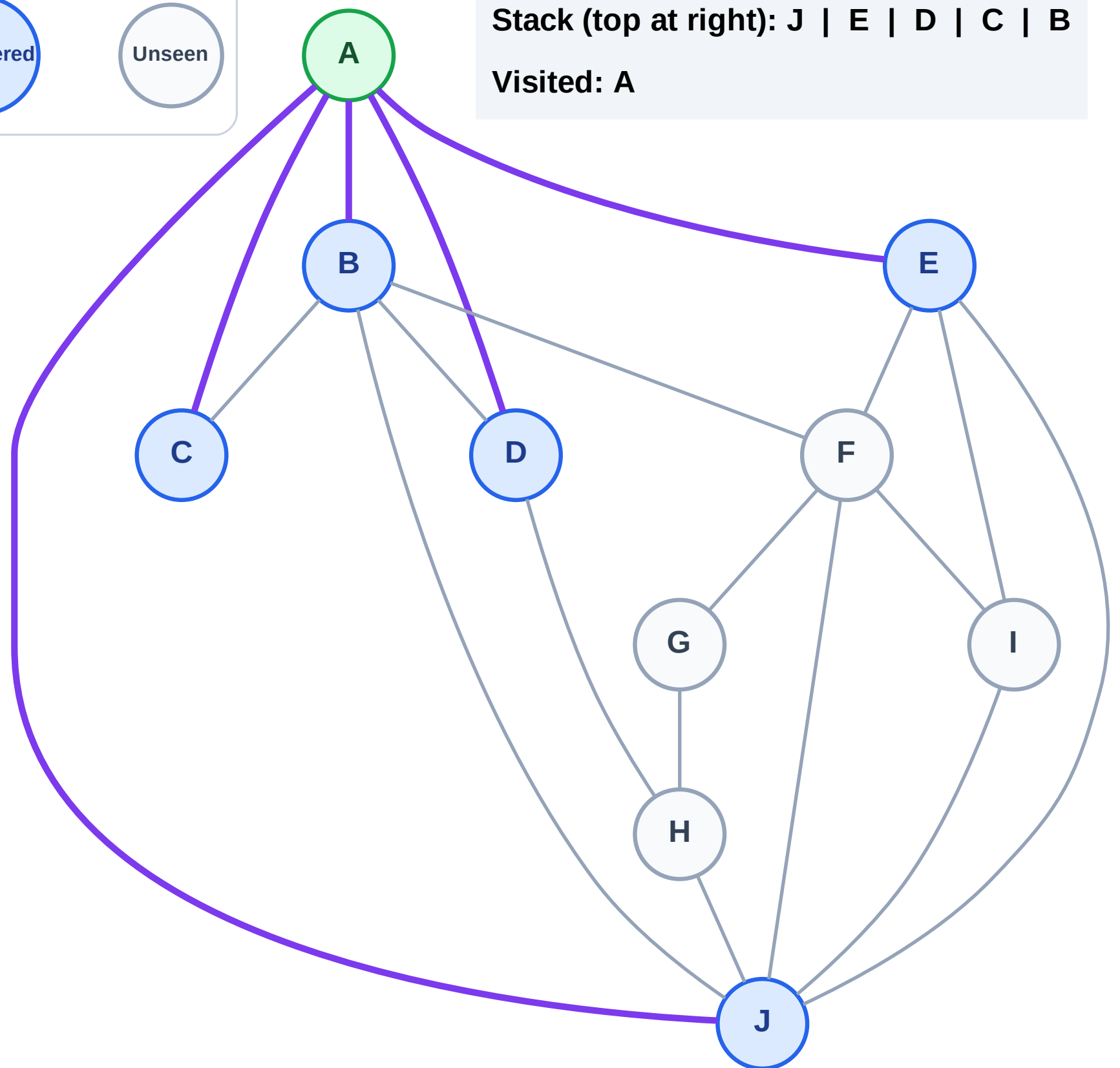
Step 3: Discover J, E, D, C, B from A

Legend



Stack (top at right): J | E | D | C | B

Visited: A



DFS Traversal

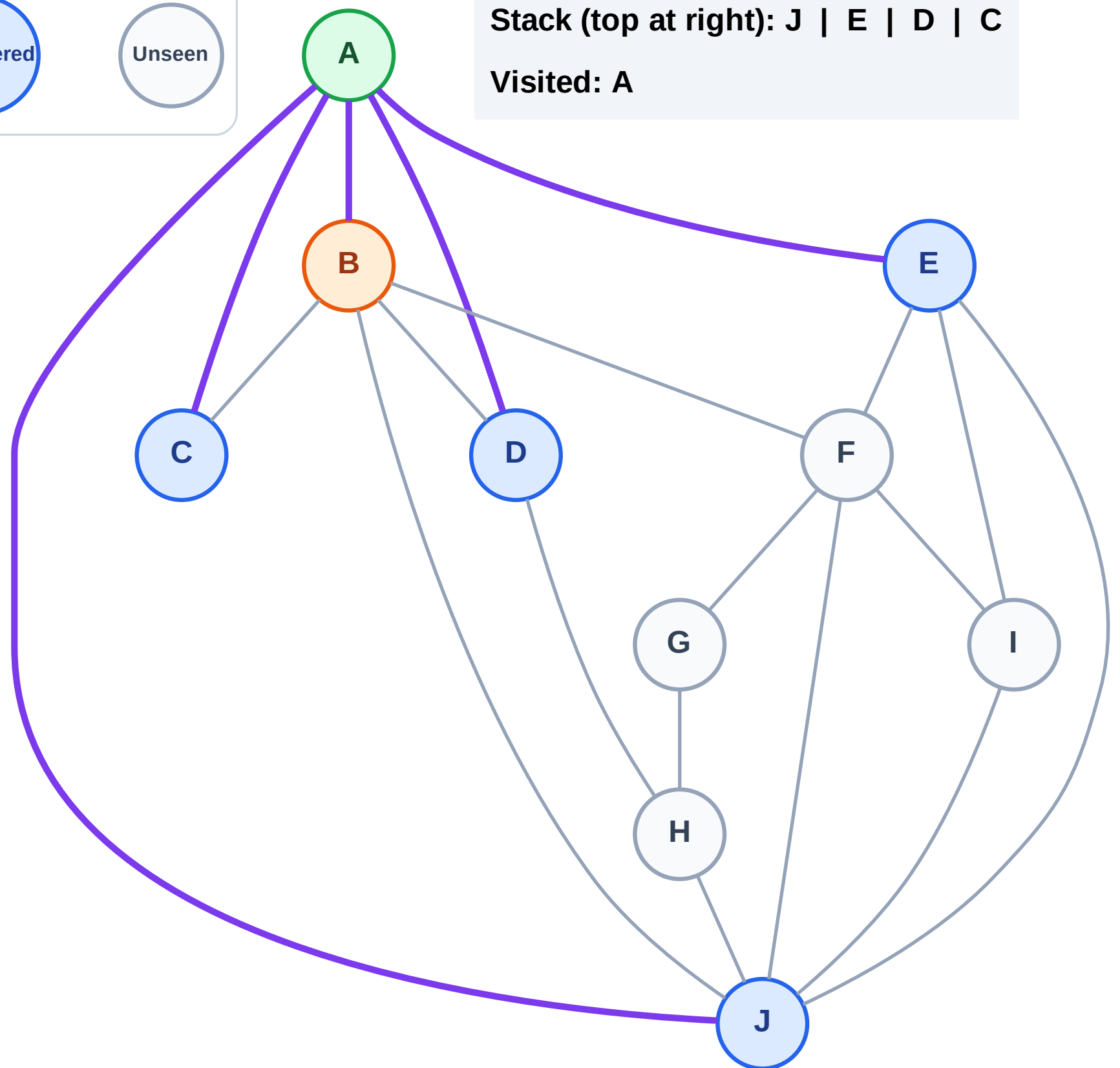
Step 4: Process node B

Legend



Stack (top at right): J | E | D | C

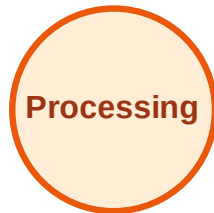
Visited: A



DFS Traversal

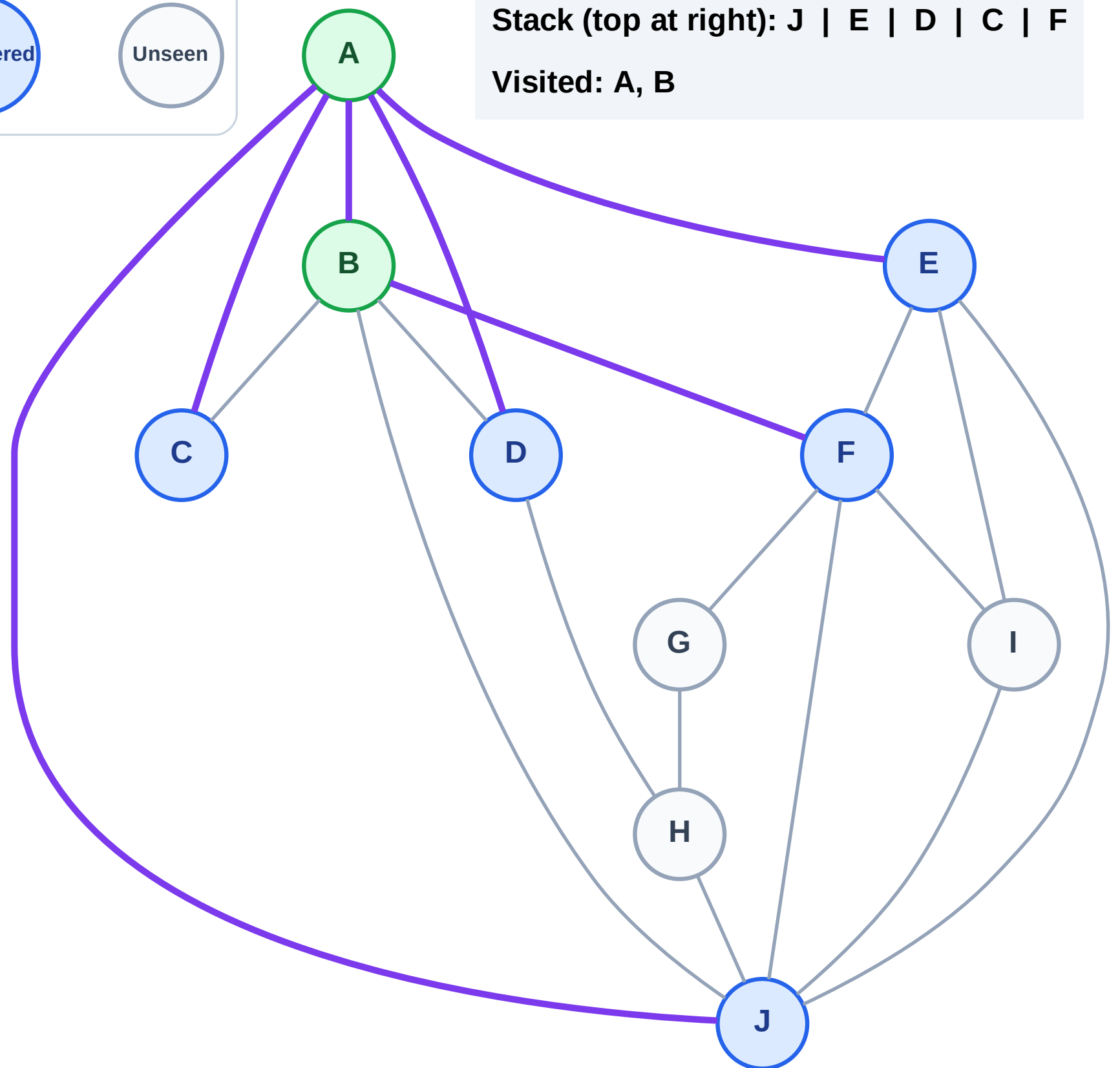
Step 5: Discover F from B

Legend



Stack (top at right): J | E | D | C | F

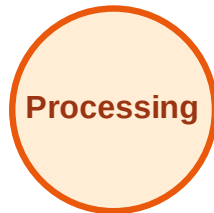
Visited: A, B



DFS Traversal

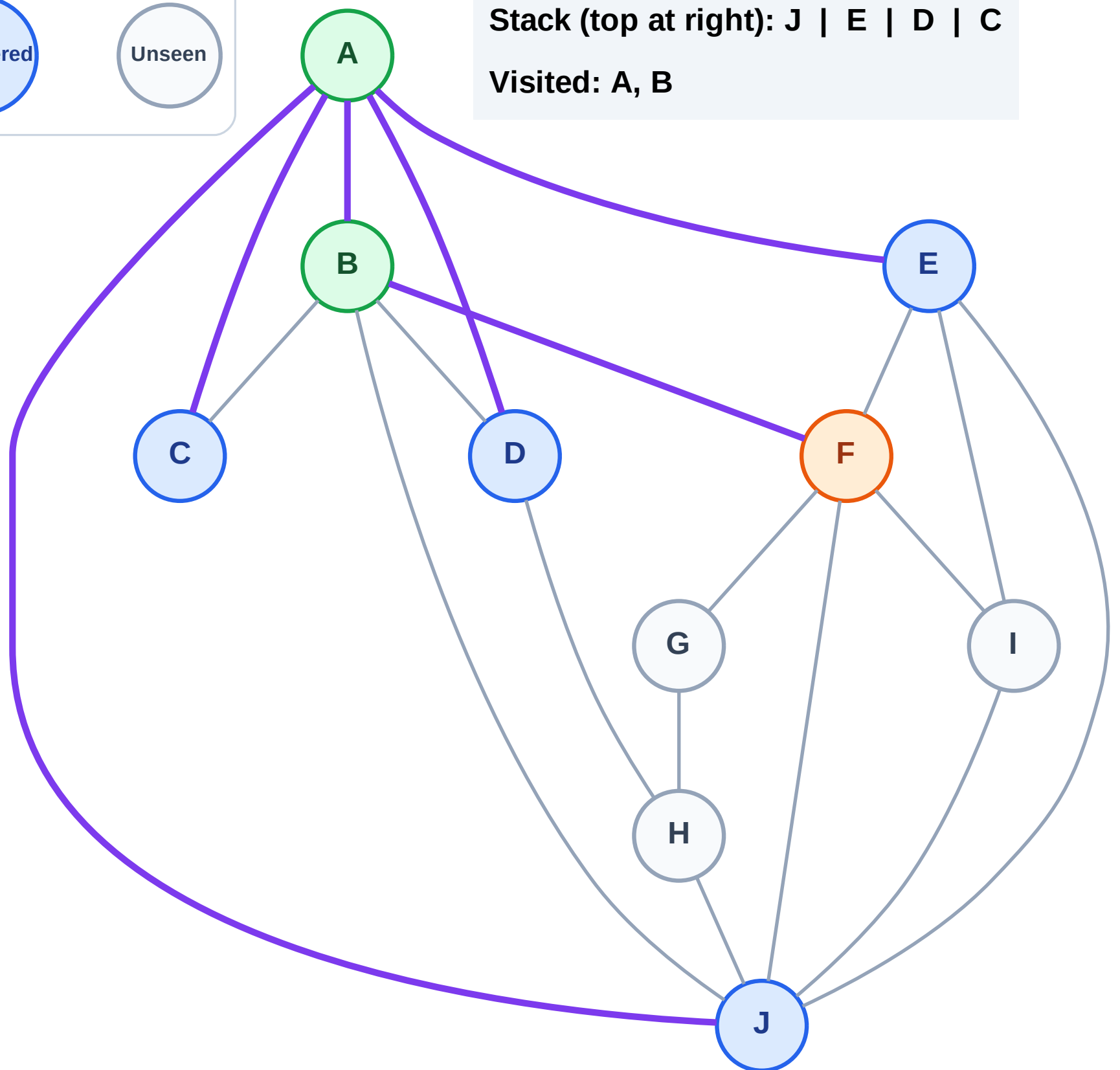
Step 6: Process node F

Legend



Stack (top at right): J | E | D | C

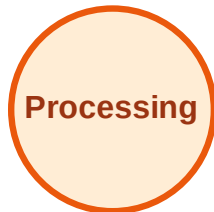
Visited: A, B



DFS Traversal

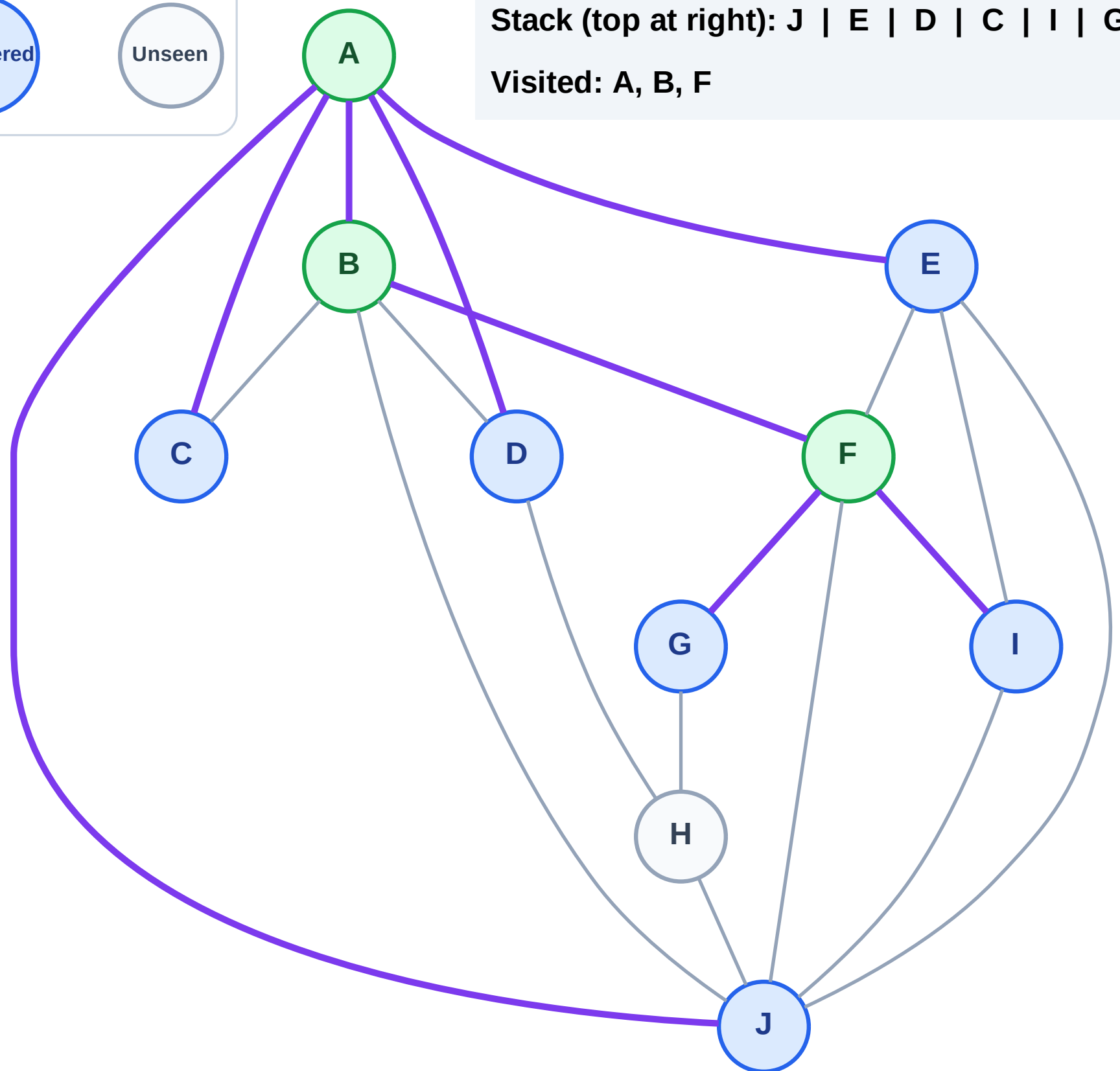
Step 7: Discover I, G from F

Legend



Stack (top at right): J | E | D | C | I | G

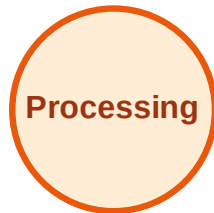
Visited: A, B, F



DFS Traversal

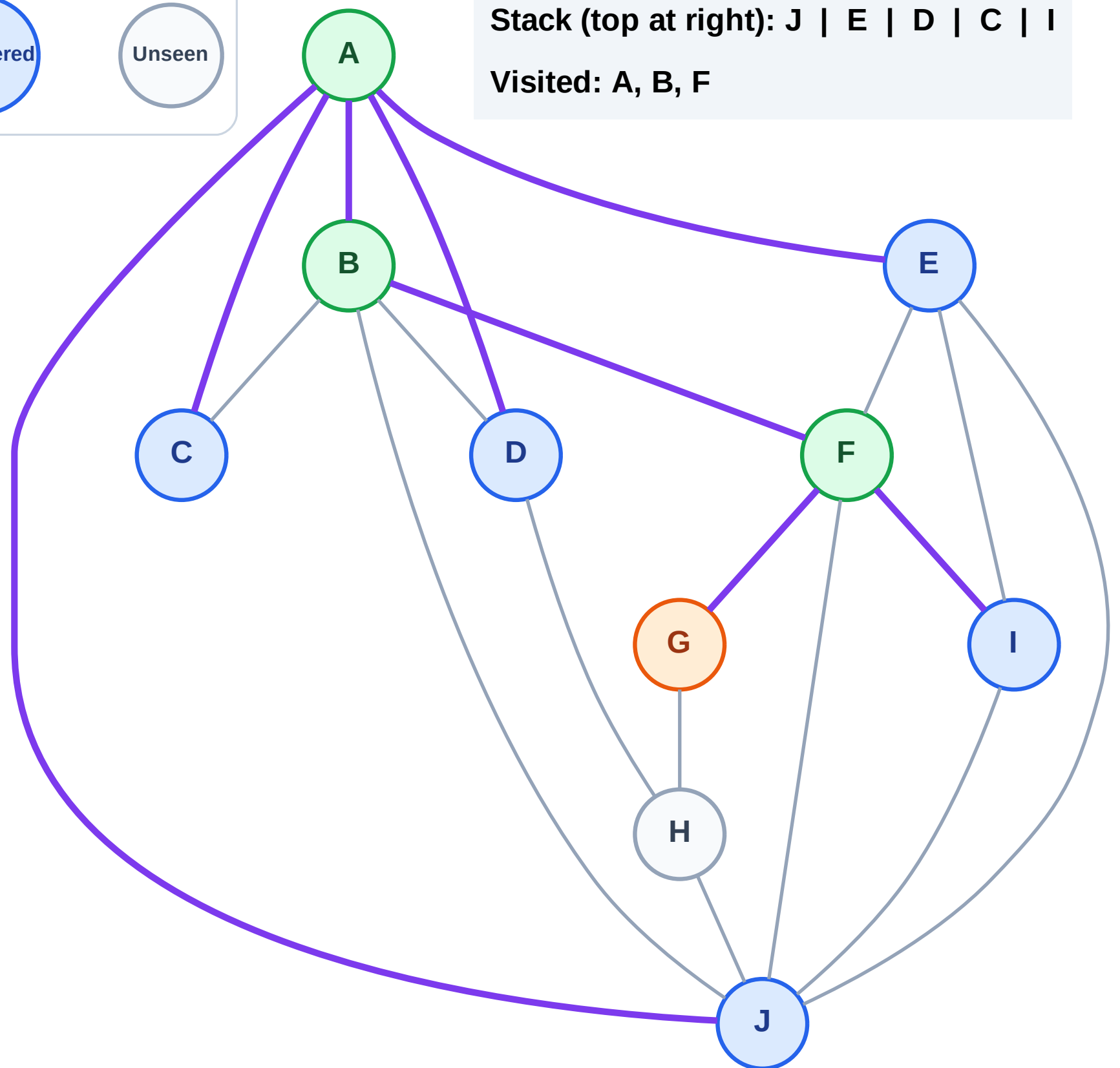
Step 8: Process node G

Legend



Stack (top at right): J | E | D | C | I

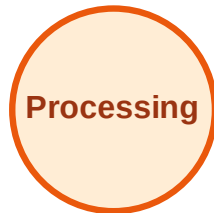
Visited: A, B, F



DFS Traversal

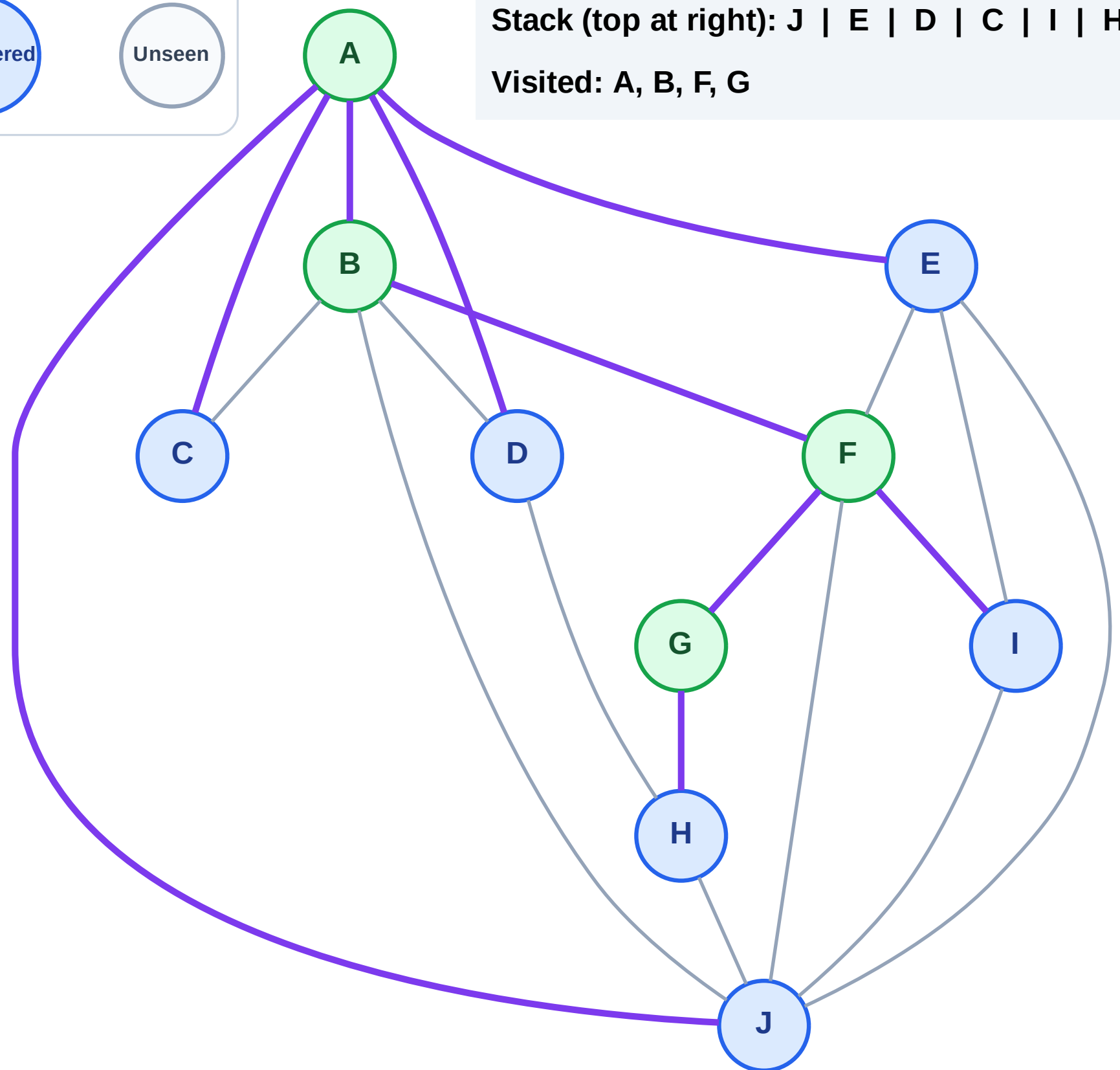
Step 9: Discover H from G

Legend



Stack (top at right): J | E | D | C | I | H

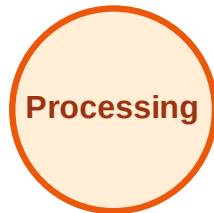
Visited: A, B, F, G



DFS Traversal

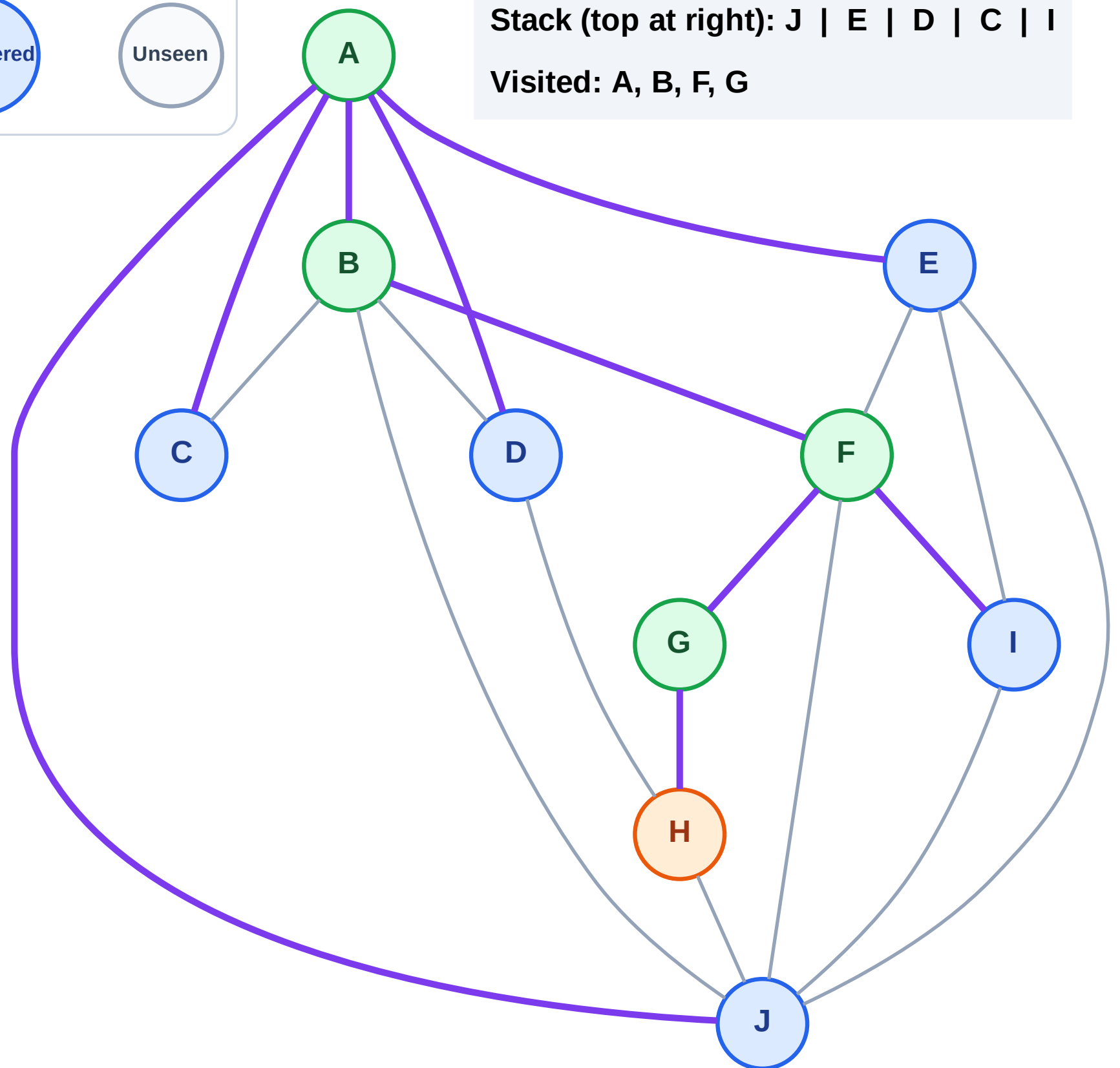
Step 10: Process node H

Legend



Stack (top at right): J | E | D | C | I

Visited: A, B, F, G



DFS Traversal

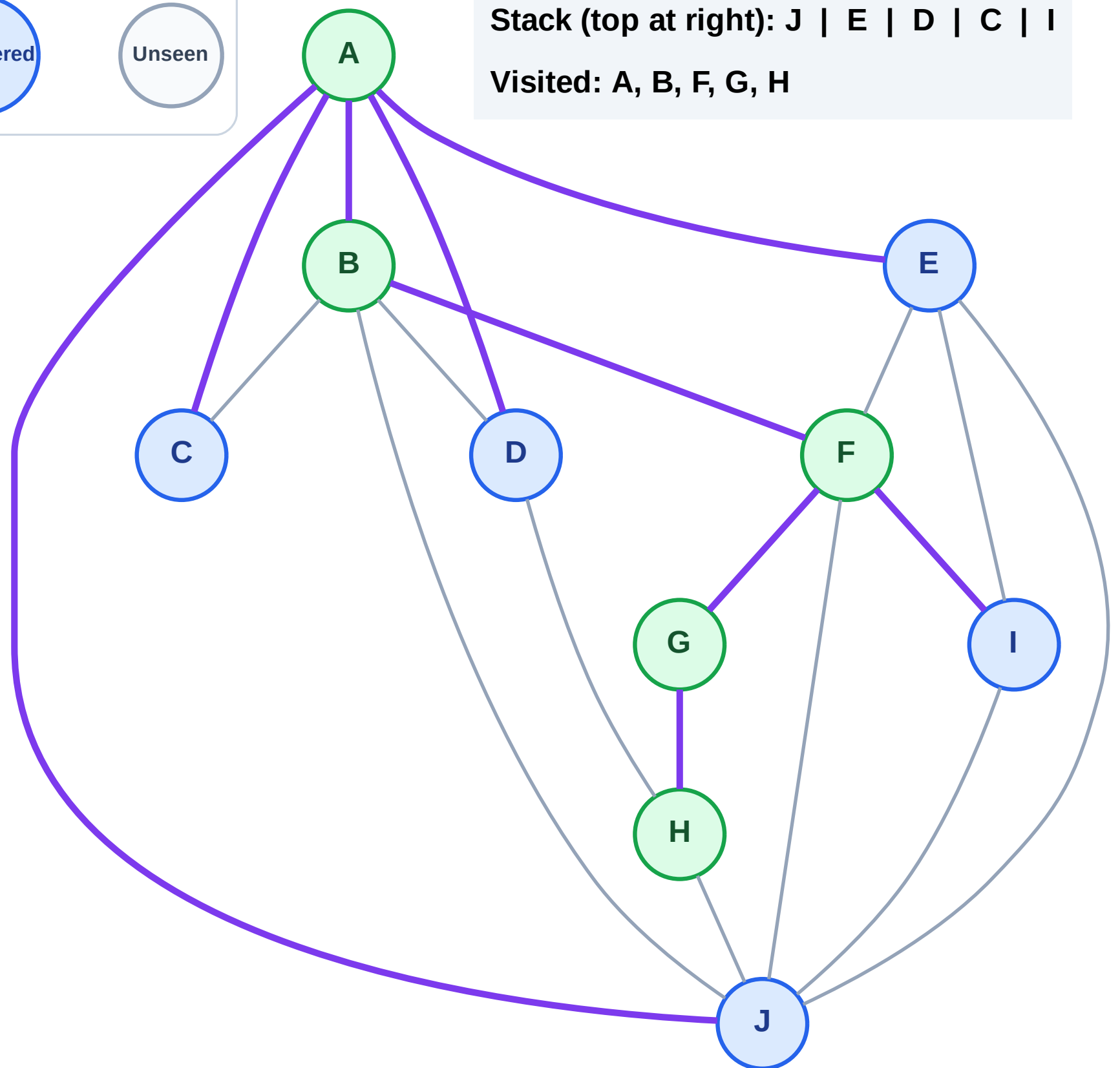
Step 11: No new neighbors from H

Legend



Stack (top at right): J | E | D | C | I

Visited: A, B, F, G, H



DFS Traversal

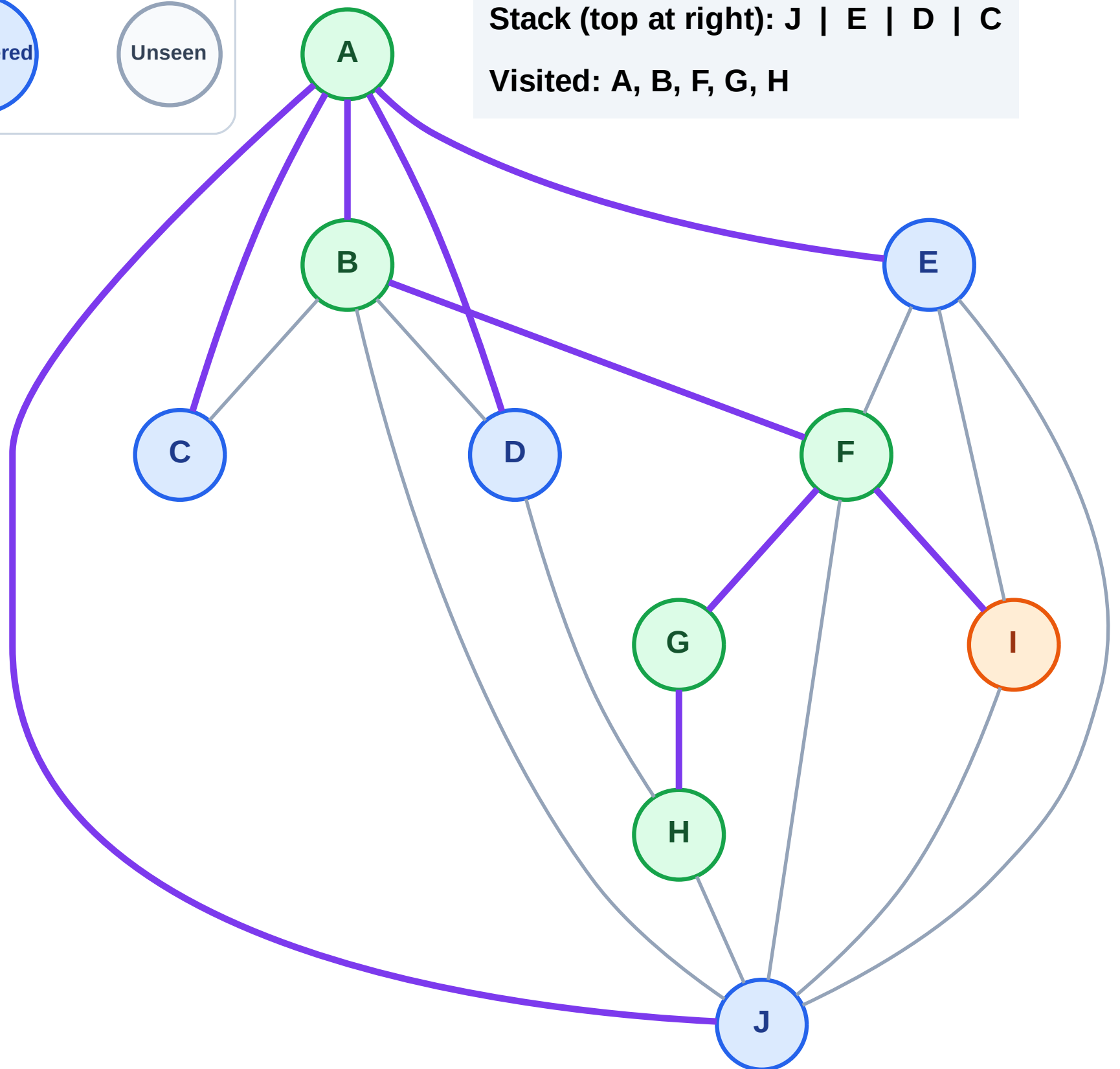
Step 12: Process node I

Legend



Stack (top at right): J | E | D | C

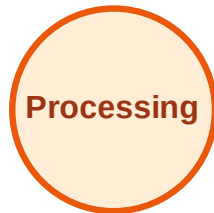
Visited: A, B, F, G, H



DFS Traversal

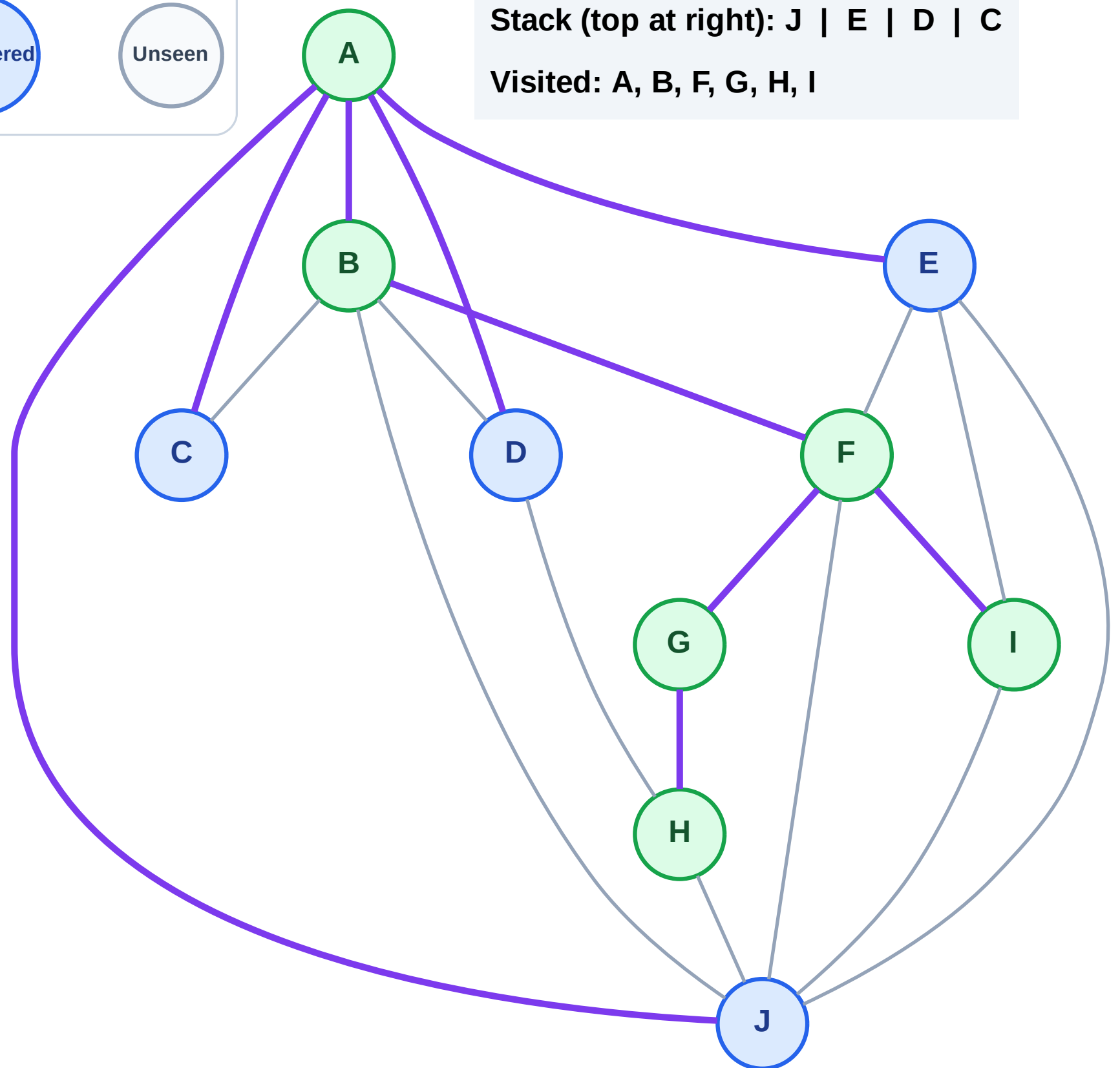
Step 13: No new neighbors from I

Legend



Stack (top at right): J | E | D | C

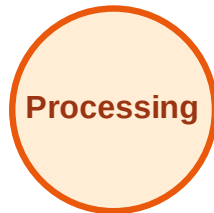
Visited: A, B, F, G, H, I



DFS Traversal

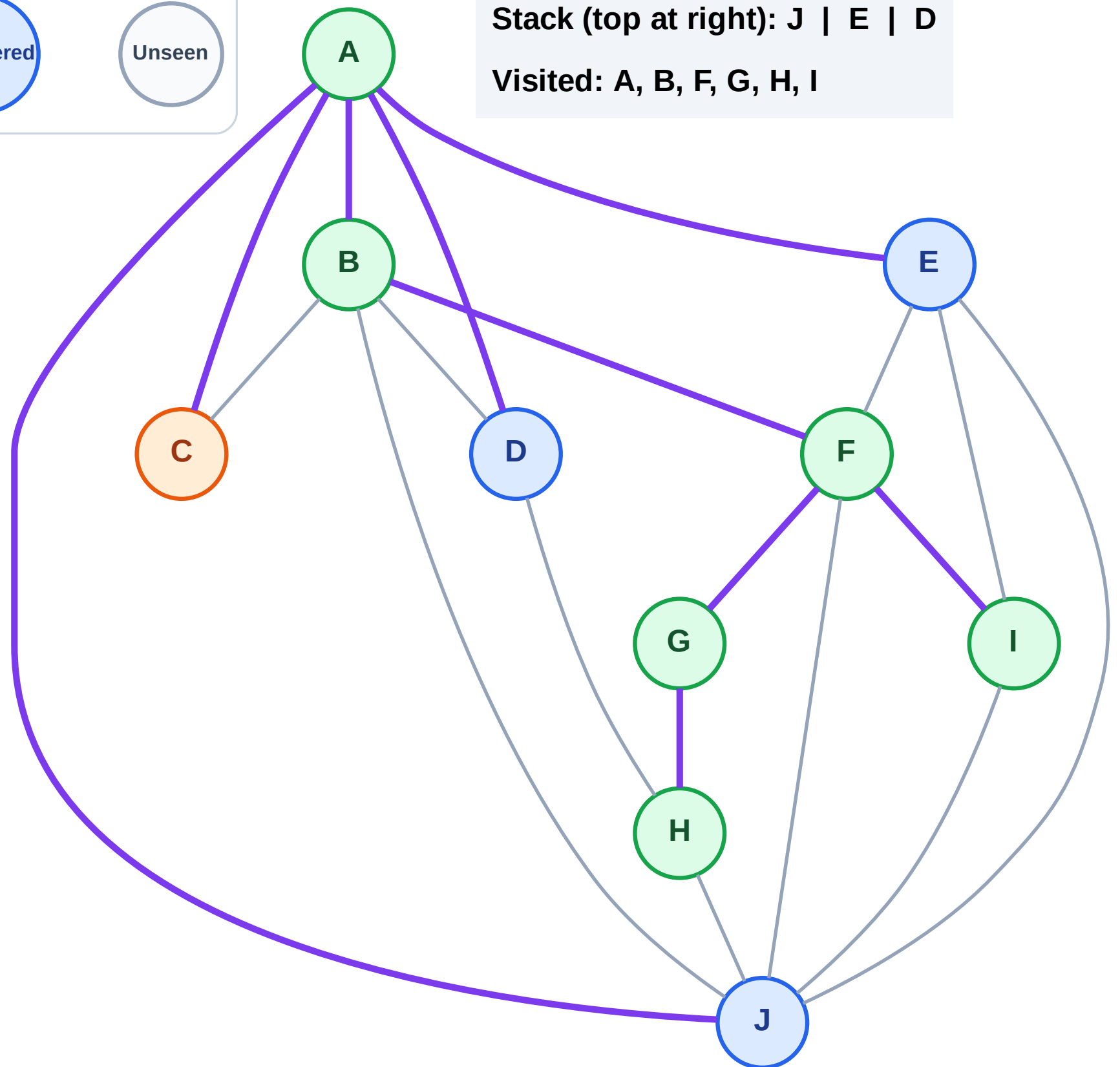
Step 14: Process node C

Legend



Stack (top at right): J | E | D

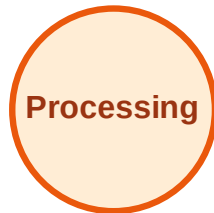
Visited: A, B, F, G, H, I



DFS Traversal

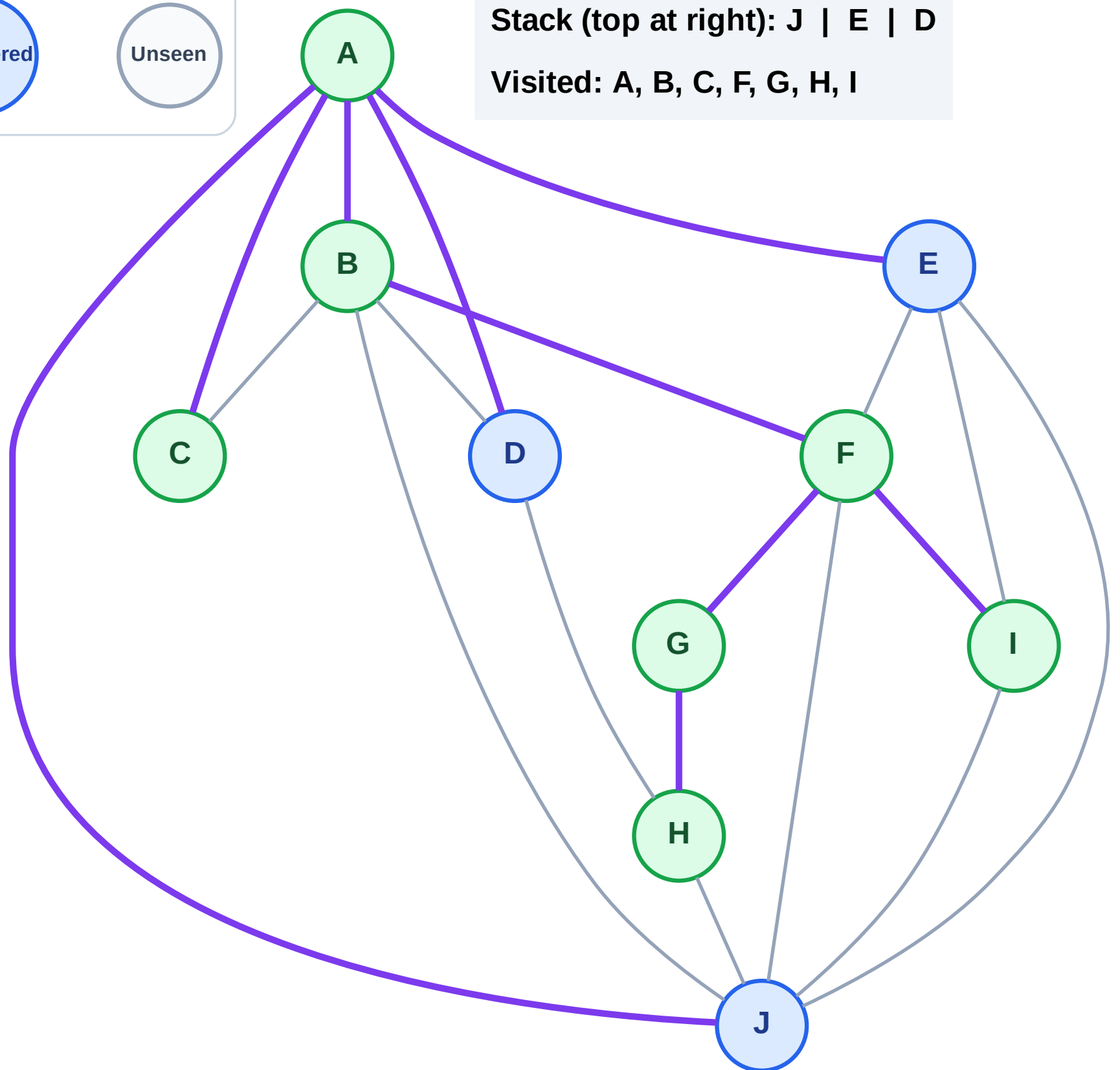
Step 15: No new neighbors from C

Legend



Stack (top at right): J | E | D

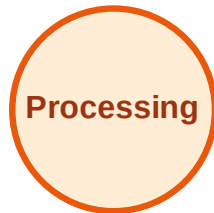
Visited: A, B, C, F, G, H, I



DFS Traversal

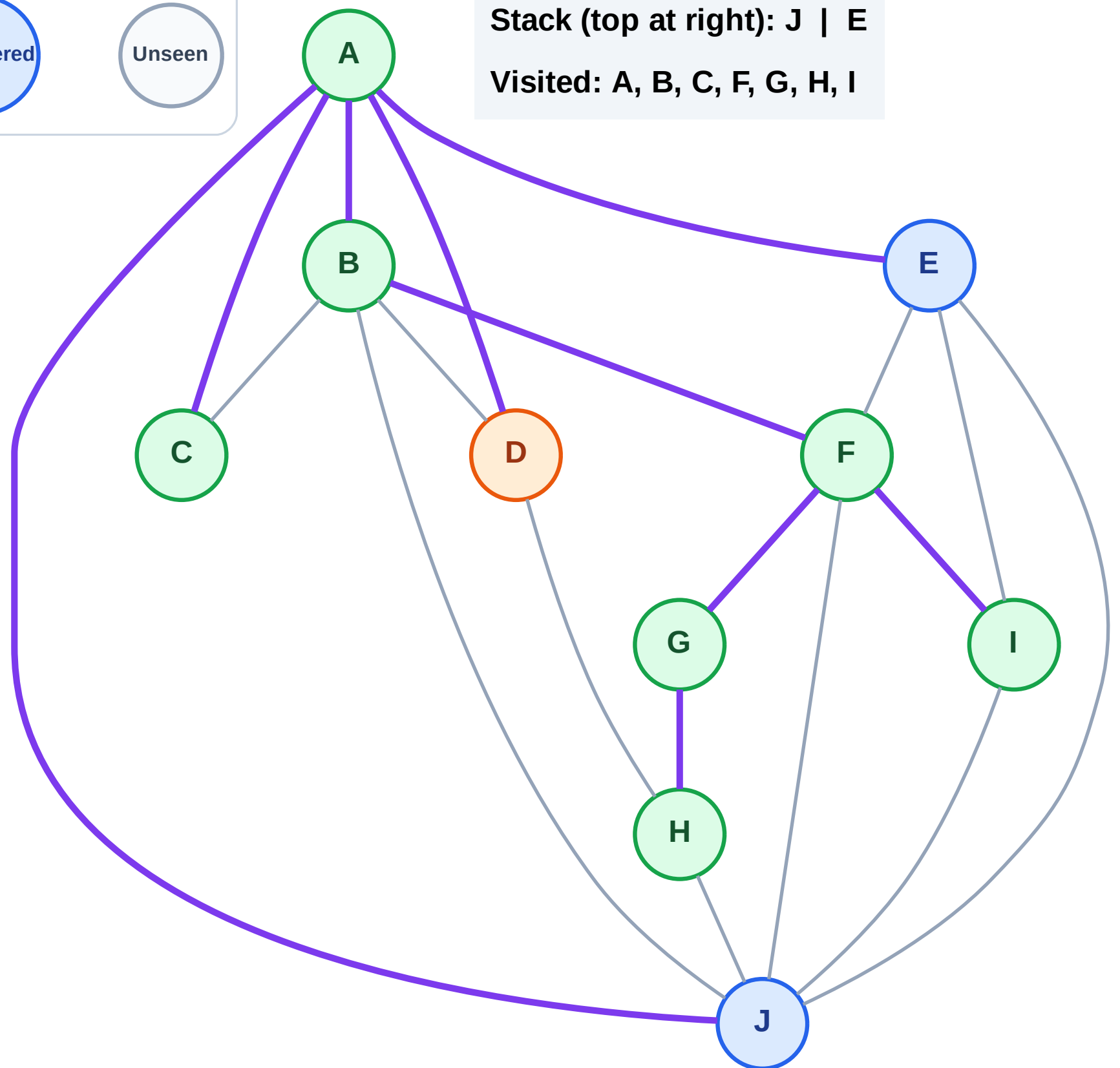
Step 16: Process node D

Legend



Stack (top at right): J | E

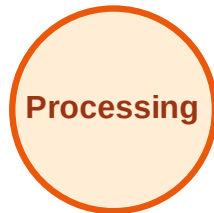
Visited: A, B, C, F, G, H, I



DFS Traversal

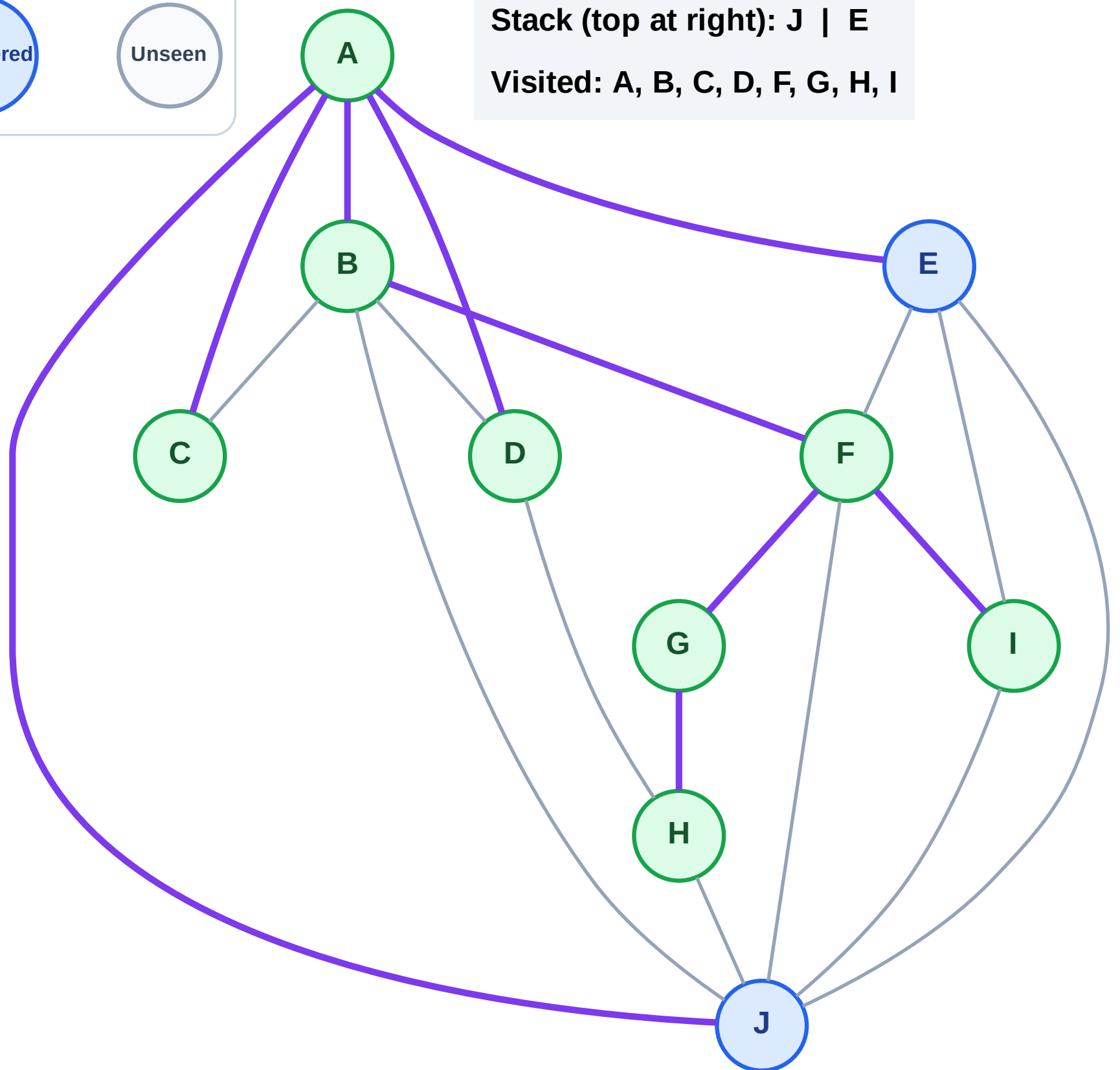
Step 17: No new neighbors from D

Legend



Stack (top at right): J | E

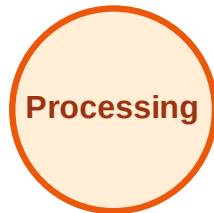
Visited: A, B, C, D, F, G, H, I



DFS Traversal

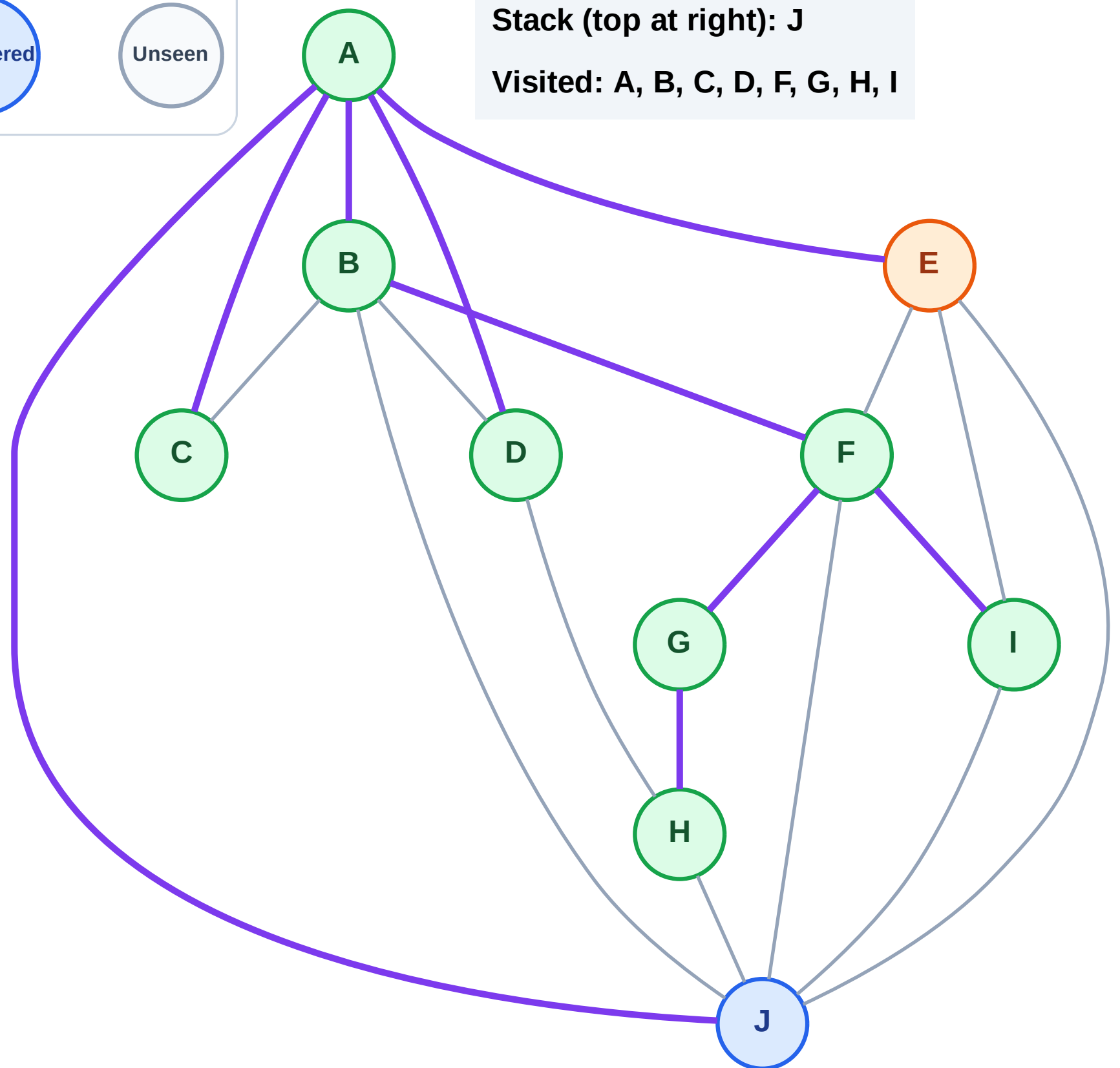
Step 18: Process node E

Legend



Stack (top at right): J

Visited: A, B, C, D, F, G, H, I



DFS Traversal

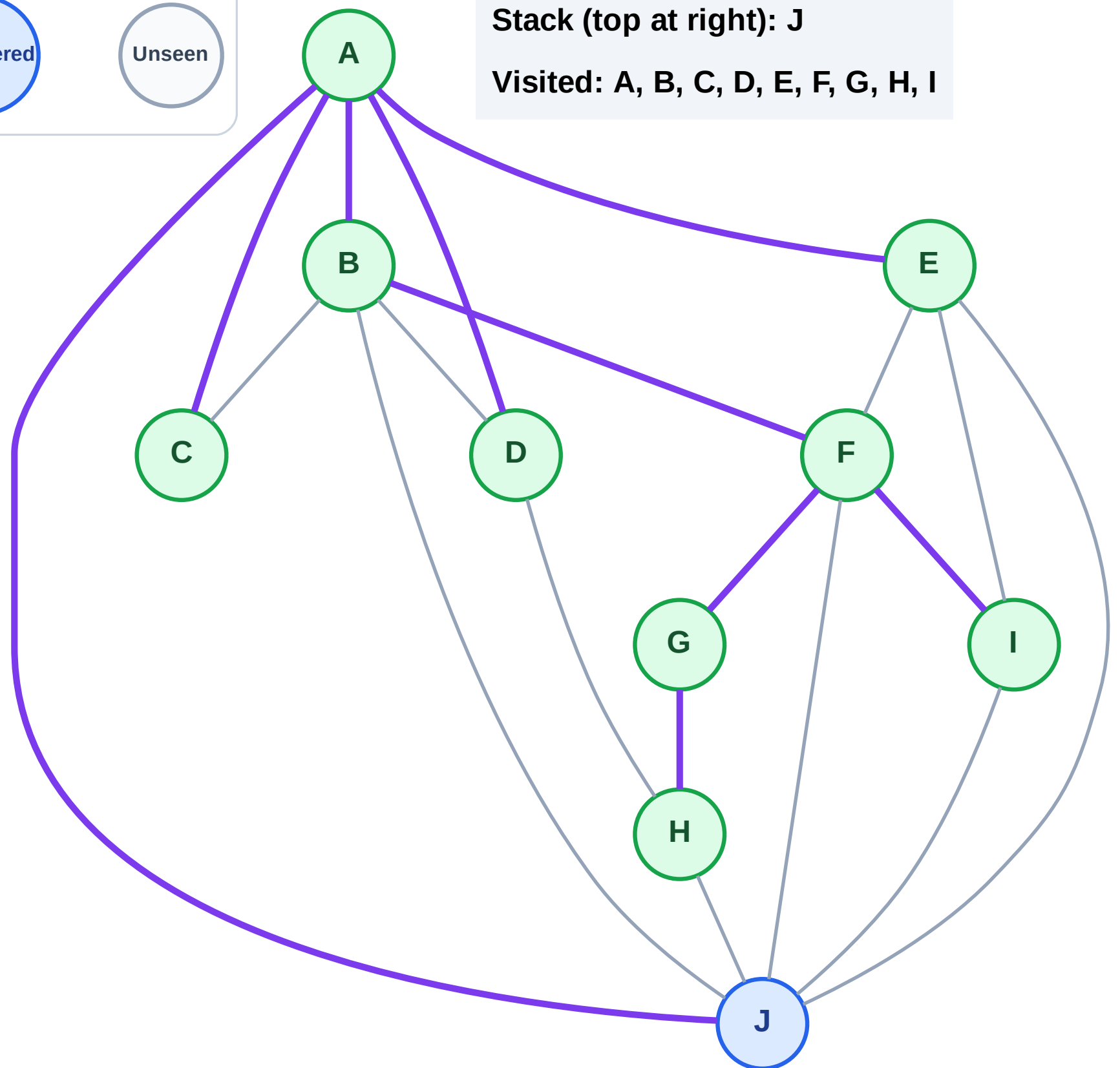
Step 19: No new neighbors from E

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

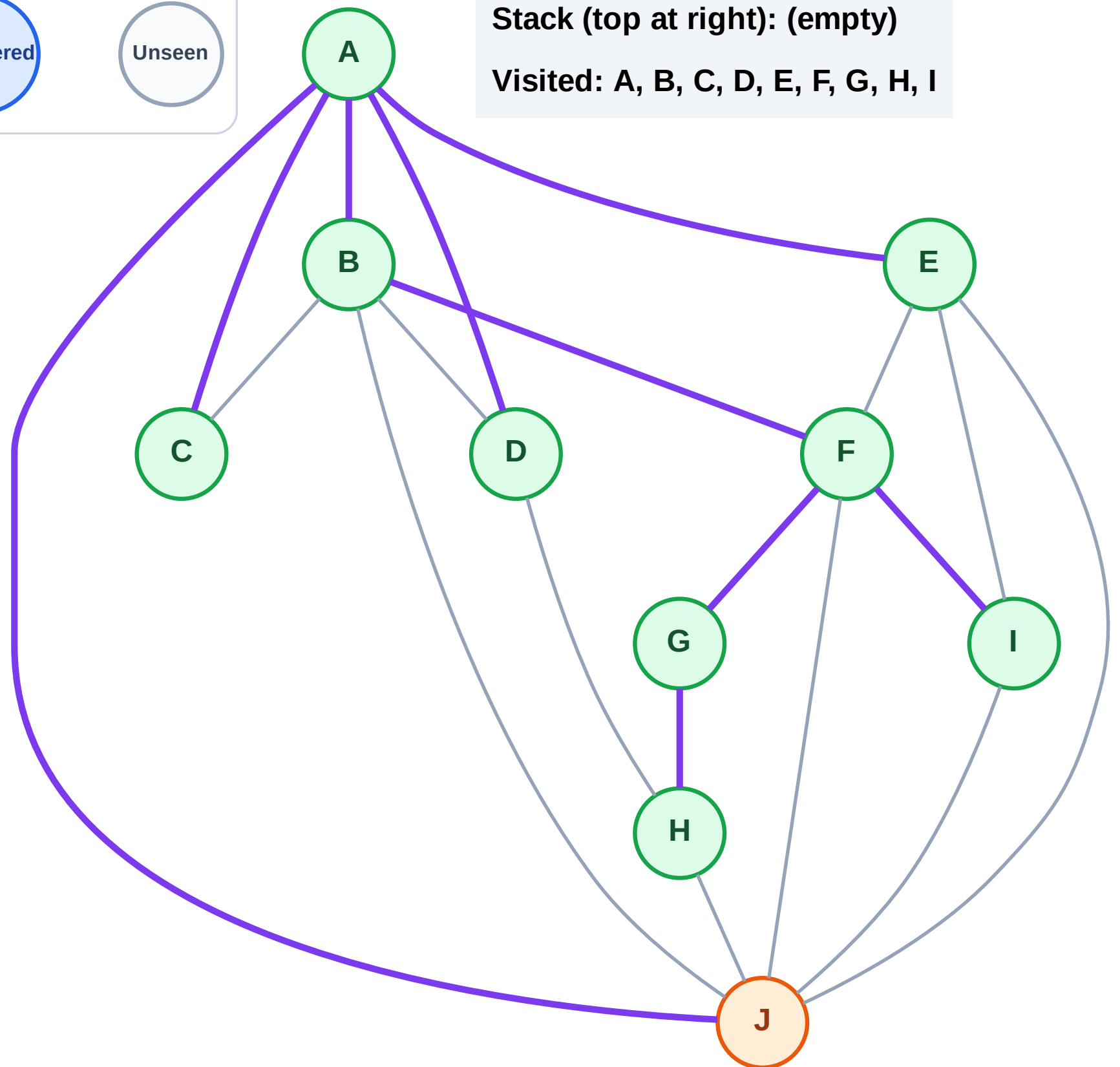
Step 20: Process node J

Legend



Stack (top at right): (empty)

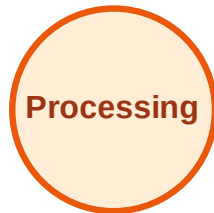
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

